GP-RareLink interim report

GP-RareLink: literature-curated, database-validated genotype–phenotype records for six rare diseases

Interim report on all work to date (internal review draft)

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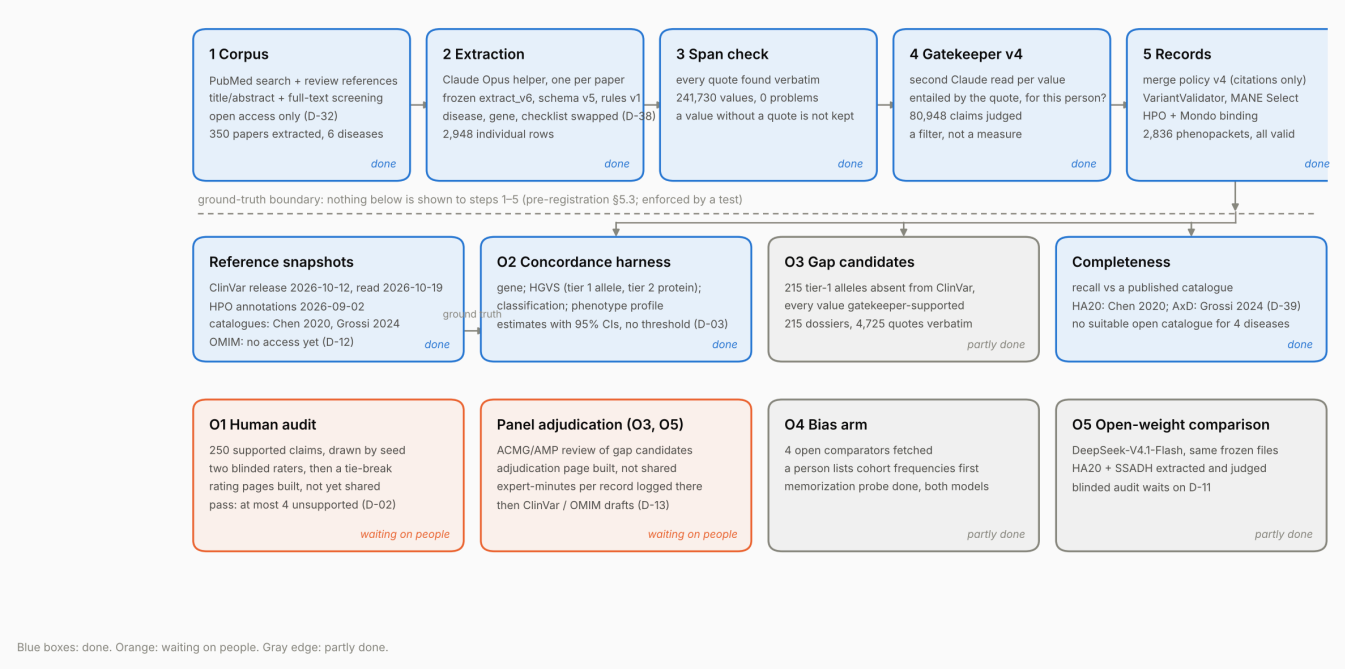


Figure 1. The pipeline, the ground-truth boundary, the harness and the human steps, with each part's status.

Status at a glance (2026-10-28; repository commit `a5e07fc`)

- **Done with the frozen pipeline (`pipeline-v1.0`, pre-registration v1.0 frozen 2026-10-20):** extraction, span check and gatekeeper on 350 open-access papers across six diseases; 2,836 schema-valid GA4GH phenopackets; a 10% run-to-run repeat; concordance against a dated ClinVar snapshot; 215 gap-variant candidates with evidence dossiers; completeness against two published catalogues; memorization probes; the Claude vs open-weight comparison on two diseases.
- **Not yet done, and needing people:** the pre-registered 250-claim human audit (O1, the one confirmatory outcome) is drawn and its blinded rating pages are built, but nobody has rated a claim; panel adjudication of gap candidates (O3) has not started; the bias arm (O4) waits on a person listing cohort frequencies; OMIM has not been checked (no access, D-12).
- **Headline numbers so far** (five confirmatory diseases; six-disease figures in brackets): 265 [350] papers, 2,067 [2,836] deduplicated individuals, 1,235 [1,793] with a target-gene variant; gatekeeper v4 judged 0.43% (248/57,819; exact 95% CI 0.38–0.49) of model claims unsupported [0.36% (294/80,948; exact 95% CI 0.32–0.41)]; 151 [215] gap candidates absent from ClinVar.
- **The one caveat that governs everything else:** the gatekeeper's rate is a filter's reading, not an error rate. In the pilot, humans found 6% unsupported where the gatekeeper found 0.04–0.3%. Until O1 is rated, no accuracy claim can be made for the frozen pipeline.
- **Spend:** about $1,861 of $45,000 by this session's own cost readings (estimates, not billing figures), plus $16–20 for the open-weight model billed outside the award.

Abstract

Background. Causative variants and phenotypes for rare diseases are published in case reports faster than they reach ClinVar, OMIM or structured phenotype resources. GP-RareLink asks whether an LLM-based extraction pipeline, with ClinVar and OMIM held out as the answer key, can build provenance-first genotype–phenotype records at a provable accuracy and at a scale that closes part of that gap. This interim report covers everything done between project start (2026-09-21) and 2026-10-28, for internal review. It is not a results paper: the one confirmatory outcome has not been measured yet.

Methods. Six diseases (*TNFAIP3*, *ALDH5A1*, *ABCA1*, *ITGB2*, *GFAP*, *LAMP2*). A pre-registered pipeline, frozen on 2026-10-20 after a one-disease pilot on A20 haploinsufficiency (HA20), extracted per-individual tables from open-access papers with Claude Opus helper agents, checked every quoted span verbatim by script, judged every value with a second model read (gatekeeper v4), merged re-reported individuals on citations only, normalized variants with VariantValidator, bound phenotypes to HPO and built GA4GH phenopackets. A harness that extraction never sees compared the records with a dated ClinVar release and two published variant catalogues, and listed variants absent from ClinVar. A 250-claim audit sample was drawn by seed for two blinded panel raters. The same files were run on an open-weight model (DeepSeek-V4.1-Flash) for two diseases.

Results. 350 papers yielded 2,948 individual rows, merged to 2,836 identities and 2,836 valid phenopackets (1,793 with a target-gene variant; 12,679 observed and 3,371 excluded HPO terms). Every one of 241,730 quoted values was found verbatim in its paper. Gatekeeper v4 judged 0.43% (248/57,819; exact 95% CI 0.38–0.49) of model claims unsupported across the five confirmatory diseases; with HA20 included, 0.36% (294/80,948; exact 95% CI 0.32–0.41). Unsupported claims cluster in a few papers, and the commonest cause is a relative's or a cited patient's variant field quoted from the proband's sentence. A 10% repeat agreed on 97–100% of checklist cells and 0.83–1.00 of variant keys. Against ClinVar's own citations, HGVS recall at tiers 1+2 ranged from 15% to 67% by disease, with misses concentrated in a handful of papers; 35–87% of extracted tier-1 alleles were present in ClinVar; classification agreement where authors state one was 59–92%. Completeness against published catalogues was 9/9 (HA20) and 54/56 (AxD) restricted variants. 215 tier-1 alleles with gatekeeper-supported values were absent from ClinVar in two successive releases (151 outside HA20), and 53 supported variants sit in ClinVar only as VUS or conflicting. Neither model showed memorization of the papers. The open-weight model produced largely the same variants at about 1/15 the cost per paper and 7× the wall-clock, but left 3–4% of values without a quote and was judged unsupported on 8.6% (HA20) and 6.1% (SSADH) of claims by the same Claude gatekeeper, against 0.20% and 0.38% for Claude.

Gaps. The primary outcome (O1) is unmeasured because the panel has not rated; the pilot's two 50-claim audits found 6.0% and 6.3% unsupported, above the 2% target, and each exposed an error class that the frozen pipeline now guards against but has not been re-audited for. OMIM is absent from every concordance and gap figure. Open-access-only corpora hold 14–53% of the eligible literature. The bias arm has no numbers. A costed plan to close each gap is in §5.

Conclusion. The pipeline runs end to end on six diseases with complete provenance and stable repeats, and it finds a plausible set of database gaps. Whether its values are right at the pre-registered level is the open question, and it is a question only the human audit can answer.

1. Introduction

Variant and phenotype knowledge for rare diseases is split across ClinVar, OMIM, HPO annotations, Monarch's phenopacket-store and, mostly, individual case reports. Many causative variants published in case reports never reach a database: for Alexander disease, a 2024 review counts 182 causative variants across 550 patients against 76 pathogenic or likely-pathogenic ClinVar records (release 2026-10-12). A lab querying ClinVar today cannot see a variant published in 2011 that was never submitted.

Research question. Can the SLIDERS method, powered by Claude, build consolidated variant ↔ phenotype ↔ disease records from published case reports at an accuracy we can prove and at a scale that materially closes this gap?

Design. Claude reads case reports for six rare diseases and emits GA4GH Phenopackets with a PMID and verbatim span on every value. ClinVar and OMIM are never shown to the extraction step; they are the answer key. Five pre-registered outcomes carry the study (Table 1). Only O1 has a pass/fail rule. Headline pooled figures use the five diseases other than HA20, which was used to develop the pipeline (D-18); figures that include HA20 are secondary.

Table 1. Pre-registered outcomes and their status.

#	Outcome	Type	Pre-specified target	Status at 2026-10-28
O1	Unsupported-value rate, human audit of 250 claims	primary, confirmatory	observed rate < 2%, i.e. at most 4 of 250 (rule C, D-02)	sample drawn (seed 40250); rating pages built; **no claim rated yet**
O2	Concordance with ClinVar and OMIM on gene, HGVS, classification, phenotype	primary, estimation	none; estimates with CIs (D-03)	ClinVar half done on all six diseases; **OMIM not assessed** (D-12)
O3	Gap closure: variants absent from both databases, adjudicated, submitted	primary, descriptive	none; counts by stage	215 ClinVar-only candidates with dossiers; **adjudication not started**

#	Outcome	Type	Pre-specified target	Status at 2026-10-28
O4	Ascertainment bias vs natural-history cohorts; memorization probe	secondary	none	memorization probe done (both models); **frequency deviation waits on a person**
O5	Efficiency: expert-minutes per accepted record; Claude vs open-weight model	secondary	none	open-weight run and gatekeeper done on 2 diseases; **blinded audit and expert-minutes not measured**

What this document is. An interim report for internal review, generated by script (`analysis/interim_paper/build.py`) from the committed outputs at commit `a5e07fc`, so that every number traces to a file and a date. It follows the pre-registration's reporting rules: per-disease results, intervals on proportions, HA20 flagged throughout, deviations listed (§2.9), and nothing called "high" or "accurate" without its interval and definition beside it. It is not for distribution outside the team.

2. Methods

The frozen plan (`preregistration/analysis_plan.md`, v1.0, 2026-10-20; tags `prereg-v1.0` and `pipeline-v1.0` on commit `ef3133e`) is the method of record. This section summarizes it and names what was built.

2.1 Diseases, genes and reference counts

Table 2. The six diseases. ClinVar counts are distinct VariationIDs on single-gene GRCh38 records in the release of 2026-10-12 (read 2026-10-19); P/LP sums Pathogenic, Likely pathogenic and Pathogenic/Likely pathogenic. OMIM allelic-variant counts are not available (D-12).

Disease (gene)	OMIM	Inheritance	System	ClinVar, single-gene records	MANE Select	Mondo
A20 haploinsufficiency (HA20, *TNFAIP3*)*	616744	AD	Auto-inflammatory	764 records; 99 P/LP; 375 VUS; 27 conflicting	NM_001270508.2	MONDO:0800045
SSADH deficiency (SSADH, *ALDH5A1*)	271980	AR	Neurometabolic	914 records; 152 P/LP; 333 VUS; 53 conflicting	NM_001080.3	MONDO:0010083
Tangier disease (Tangier, *ABCA1*)	205400	AR	Lipid metabolism	2,006 records; 85 P/LP; 705 VUS; 141 conflicting	NM_005502.4	MONDO:0008783
Leukocyte adhesion deficiency I (LAD-I, *ITGB2*)	116920	AR	Immunology	949 records; 87 P/LP; 345 VUS; 49 conflicting	NM_000211.5	MONDO:0007293
Alexander disease (AxD, *GFAP*)	203450	AD / de novo	Neurology	627 records; 76 P/LP; 262 VUS; 36 conflicting	NM_002055.5	MONDO:0008752
Danon disease (Danon, *LAMP2*)	300257	X-linked dominant	Cardiac / neuromuscular	986 records; 154 P/LP; 292 VUS; 66 conflicting	NM_002294.3	MONDO:0010281

*Development disease: every extraction rule was tuned on it during the pilot (§2.8), so its results flatter the pipeline (D-18).

2.2 Corpus

- **Search.** PubMed from inception to a recorded search date (HA20 2026-10-18; the others 2026-10-19), gene symbol plus disease names and synonyms, plus backward citation searching of the key reviews through Europe PMC's reference index. No ClinVar or OMIM citation lists were used as seeds for the five new diseases (D-19). HA20's pilot corpus kept its ClinVar-seeded papers, tagged, and every O2 figure is reported with and without ClinVar-cited papers.
- **Screening.** Title/abstract screening by Claude Code helper agents with a logged reason per record, then full-text screening of every open full text with a verbatim evidence quote checked by script. A human 20-item check settled the screening calls (D-34). HA20's corpus was carried over from the pilot and was not full-text screened.
- **Eligibility.** Primary publications reporting individual-level genotype or phenotype for at least one person with the disease; reviews with a per-patient table are included and their people tagged secondary-source (D-33). Reviews without new individuals, non-English papers and papers with no individual data are excluded with the reason logged.
- **Access.** Open access only: PMC, Europe PMC and open copies found through Unpaywall (D-32). No institutional access. Supplementary files were fetched where Europe PMC served them and converted to text (capped at 60,000 characters per file).
- **Holdouts.** Natural-history comparator papers for the bias arm are marked out of the extraction corpus (AxD 2, SSADH 1, Danon 2).

2.3 Extraction

- **Model and route.** Claude Opus, as Claude Code helper agents on the funded claude.ai account (D-17, D-20): one helper per paper, reading only the paper, the instruction file, the schema and the rules file, with no web access and an instruction to use nothing it remembers about the paper. The pilot manifests record the served model ID, `claude-opus-5-5`; the six-disease manifests recorded only the helper alias "opus" at run time, and a `served_model` line stating `claude-opus-5-5` from the session configuration was added on 2026-10-28, marked as such. Instruction hashes and session cost readings are in each run manifest (`runs/pv1-<disease>/manifest.json`).

- **Frozen files.** `extract_v6.md` (instructions), `schema_<disease>_v5.json` (five tables: publications, individuals, variants, phenotypic_features, phenotype_checklist), `extraction_rules_v1.md` and `gatekeeper_v4.md`. For the five new diseases the prompt and schema were instantiated by script, swapping only the disease name, gene symbol, schema path and a 20-item checklist (D-38; 12 changed lines per disease, diffs committed).
- **What a value is.** Every asserted value carries a verbatim quote from the paper. "Not reported" is a first-class output and carries no quote. Three-state phenotypes (observed / excluded / not reported) are never collapsed. Variant fields are taken only from statements about this person (D-29); values stacked in multi-patient cells are not assigned by position (D-31); a normal finding is recorded as the abnormality it rules out, marked excluded (D-27).
- **Checklists.** Twenty disease-specific features per disease, each with the three-state status for every individual (Supplementary Table S3). HA20's came from the pilot schema; the others were written by helpers from a consensus guideline or review (SSADH 38452608; Danon 37821174; Tangier 33994407; AxD 39420046, 32779598, 40520824; LAD-I 42011429, 23207660, 33391282), every quote checked verbatim.

2.4 Span check and gatekeeper

- **Span check.** A deterministic check that every quoted span occurs verbatim in the source text, run on all values, not a sample. A helper that left quotes empty was resumed for a repair-only pass; each repair was diffed to show only the target cells changed (the manifests list them; HA20 had 6).
- **Gatekeeper v4.** A second Claude read of each model claim (claims that are not bookkeeping, such as PMIDs and repeated labels), one helper per chunk of up to 350 claims, asking whether the quote entails the value *for the person the row is about*. It records whom the quote is about (this person, a named group, the cohort, another person, the paper, unclear) and gives one of four verdicts: supported, partially supported, unsupported, span not found. No value is exempted by a script match of value and quote (D-28). Phenopackets keep only claims judged supported, and the count removed is reported. **The gatekeeper is a filter, not a measure** (§6.4 of the plan): in the pilot its own unsupported rate understated the human audit's by an order of magnitude.

2.5 Records

- **Deduplication.** Merge policy v4: merges only on an explicit citation tying a person to an earlier report, one-to-one; the pilot's sex-and-onset route was dropped after it produced all 6 false merges of 93 (D-23). A paper pair listed in `configs/duplicate_publications.csv` with evidence (the SSADH preprint and its journal version) uses a dedicated route.
- **Normalization.** VariantValidator 4.0.1 (dev), MANE Select transcript per gene read from VariantValidator on 2026-10-20 (Table 2); the original string is kept beside the normalized form; transcripts assumed or version-mapped are flagged. Only VariantValidator's normalization fields enter the pipeline; any database annotations in its responses are dropped.
- **Binding.** Checklist items map to HPO terms through per-disease maps built from the pinned HPO release (2026-09-01) and flagged for review; free-text features are bound by script first, then by helpers with an HPO lookup tool, every answer checked against the release. Disease terms come from Mondo (release 2026-10-06) by name, never by OMIM cross-reference.
- **Phenopackets.** One GA4GH Phenopacket v2 per identity, validated against the schema (D-40 per-disease settings: gene, HGNC ID, MANE transcript, Mondo term, checklist map).

2.6 Ground-truth isolation and the harness

Code under `pipeline/extract` never reads reference data; an automated test fails if any such path, loader or the tokens "clinvar", "omim" or "harness" appear there. Reference snapshots live outside the repository (`/home/user/gp-rarelink-reference/<date>/`), and only their manifests and counts are committed. The harness (`pipeline/harness/`) reads the extracted records and the snapshots:

- **ClinVar** `variant_summary.txt.gz`, release 2026-10-12, read 2026-10-19 (re-read 2026-10-19 release on 2026-10-21 for the gap re-check), restricted to single-gene GRCh38 records per gene.
- **HPO annotations** (`phenotype.hpoa`, 2026-09-02) for the phenotype-profile figure.
- **Catalogues:** Chen 2020 Table S1 for HA20 (D-30) and Grossi 2024 Supplementary Table 2 for AxD (D-39), transcribed by script. The other four candidates were dropped by the registered suitability check (no per-variant table naming sources, or not open), so those diseases report no completeness figure.
- **OMIM:** no access (D-12). Every O2 and O3 figure is ClinVar-only, and the OMIM half is "not yet assessed", never "absent".

2.7 Outcomes and statistics

- **O1.** 250 claims drawn by simple random sampling, stratified by disease in proportion to claim volume, from the model claims gatekeeper v4 judged supported (the values that survive into the records), by script with seed 40250 and before any concordance was run. Two panel members rate blind on separate pages that never show the gatekeeper's verdict; a third resolves disagreements. Pass under rule C: at most 4 unsupported of 250, reported with the exact interval and the highest rate the audit cannot exclude. A value the rater cannot verify counts as unsupported, with a sensitivity analysis without such claims.

- **O2.** Tier 1 = same GRCh38 allele (VCF key); tier 2 = same protein change for papers with no cDNA. HGVS recall = share of ClinVar's (PMID, variant) citation pairs, restricted to corpus PMIDs, that the run extracted from that PMID. Overlap = share of extracted tier-1 alleles present in ClinVar (not precision). Classification = authors' P/LP, VUS, B/LB vs ClinVar aggregate, among stated values. Phenotype profile = HPO annotation recall (term or descendant seen in ≥1 person) and the share of terms seen in ≥2 people that match an annotation. Wilson 95% intervals. No pass/fail threshold (D-03). Every miss counts as discordant; a seeded sample of 50 misses awaits human triage into five categories.
- **O3.** A gap candidate is a tier-1 allele absent from the ClinVar snapshot whose identifying values the gatekeeper judged supported in at least one row. Counted by stage: candidates, adjudicated, approved, drafted, signed off, submitted, accepted. Reclassification candidates (in ClinVar only as VUS or conflicting) are exploratory.
- **O4.** Per disease, a human lists each held-out comparator's HPO-mappable features and cohort frequencies before literature frequencies are computed; then observed ÷ (observed + excluded) per feature, with a publication-cluster bootstrap. Memorization: closed-book recall from PMID, title and year, and guided completion of a 50-word opening, on 30 papers per disease stratified by date against the model's documented cutoff (June 2026), scored against the full-text extraction.
- **O5.** Expert-minutes per accepted record, logged by the adjudication page. Claude vs DeepSeek-V4.1-Flash (D-10): same papers, launch prompt, frozen files, file tools and scoring; metrics are span-not-found rate, unsupported rate on a blinded audit of equal size (pending), HGVS concordance, coverage, cost and wall-clock.
- **Intervals.** Wilson for simple proportions; exact Clopper–Pearson for the audit and for the gatekeeper shares; pooled five-disease and six-disease figures in this report use the same forms.

2.8 The development pilot

A one-disease pilot on HA20 (2026-10-18 to 2026-10-20) built and tuned the pipeline before the freeze: two full extraction runs, a 15-paper repeat, a hand check of every merge, two 50-claim human audits by Dr. Lin, ClinVar concordance and the Chen 2020 completeness check. Its findings (§3.6) set the rules frozen as extraction v6 and gatekeeper v4. HA20 was then re-extracted from scratch with the frozen pipeline in the six-disease run; the pilot's own records are not used anywhere in this report's tables.

2.9 Deviations from the frozen plan

Three dated deviations, each approved by Dr. Lin before it took effect, are logged in `DECISIONS.md`:

- **D-38 (2026-10-20).** The frozen prompt and schema name HA20 and *TNFAIP3*; for the other five diseases they were instantiated by script (disease name, gene symbol, schema path, checklist only). Expected effect: none on HA20; the five checklists are new by design.
- **D-39 (2026-10-20).** The AxD catalogue (Grossi 2024) was dropped by the pre-extraction suitability check because Europe PMC answered HTTP 500 for its supplement; the supplement arrived after `pv1-axd` finished and was transcribed on the harness side only. Nothing was appended to the paper text extraction read.
- **D-40 (2026-10-20).** Stage 4 scripts (normalization, binding, phenopackets) read per-disease settings instead of HA20's hard-coded ones. No change to the four pinned files, metrics or thresholds.

A fourth item is a decision, not a deviation: the SSADH corpus holds one study twice (37503297, 37962671); both stay and their patients are merged with the evidence recorded.

2.10 Reproducibility

Every table and figure here is produced by `analysis/interim_paper/` from committed files (listed in Supplementary Table S1). Run manifests record the pinned files' sha256 at launch and at finish, the papers, the helpers, the repairs and the session cost readings. The reference snapshots are re-creatable by `pipeline/harness/snapshot_reference.py`, `benchmark_chen2020.py` and `catalogue_grossi2024.py`.

3. Results

3.1 Corpus

Table 3. Corpus and open-access coverage by disease. "Free elsewhere" is a Europe PMC listing of a free copy outside PMC, not fetched (an upper bound). Source: `analysis/corpus/coverage_2026-10-20.json`; run summaries.

Disease	Records screened	In corpus (title/abstract)	Open full text	+ free elsewhere (upper bound)	Eligible after full-text screening	Papers extracted	…with individual rows	ClinVar-seeded papers (D-19)	Comparator holdouts
HA20*	838	160	85 (53.1%)	60.0%	not screened (pilot corpus)	85	77	92	—
SSADH	357	171	59 (34.5%)	46.8%	40	40	40	—	30656189

Disease	Records screened	In corpus (title/abstract)	Open full text	+ free elsewhere (upper bound)	Eligible after full-text screening	Papers extracted	...with individual rows	ClinVar-seeded papers (D-19)	Comparator holdouts
Tangier	529	284	39 (13.7%)	35.6%	31	31	31	—	—
LAD-I	874	274	59 (21.5%)	36.5%	42	42	42	—	—
AxD	539	363	83 (22.9%)	38.8%	69	69	69	—	34865968, 39643430
Danon	416	244	96 (39.3%)	47.1%	83	83	83	—	21415759, 41121270

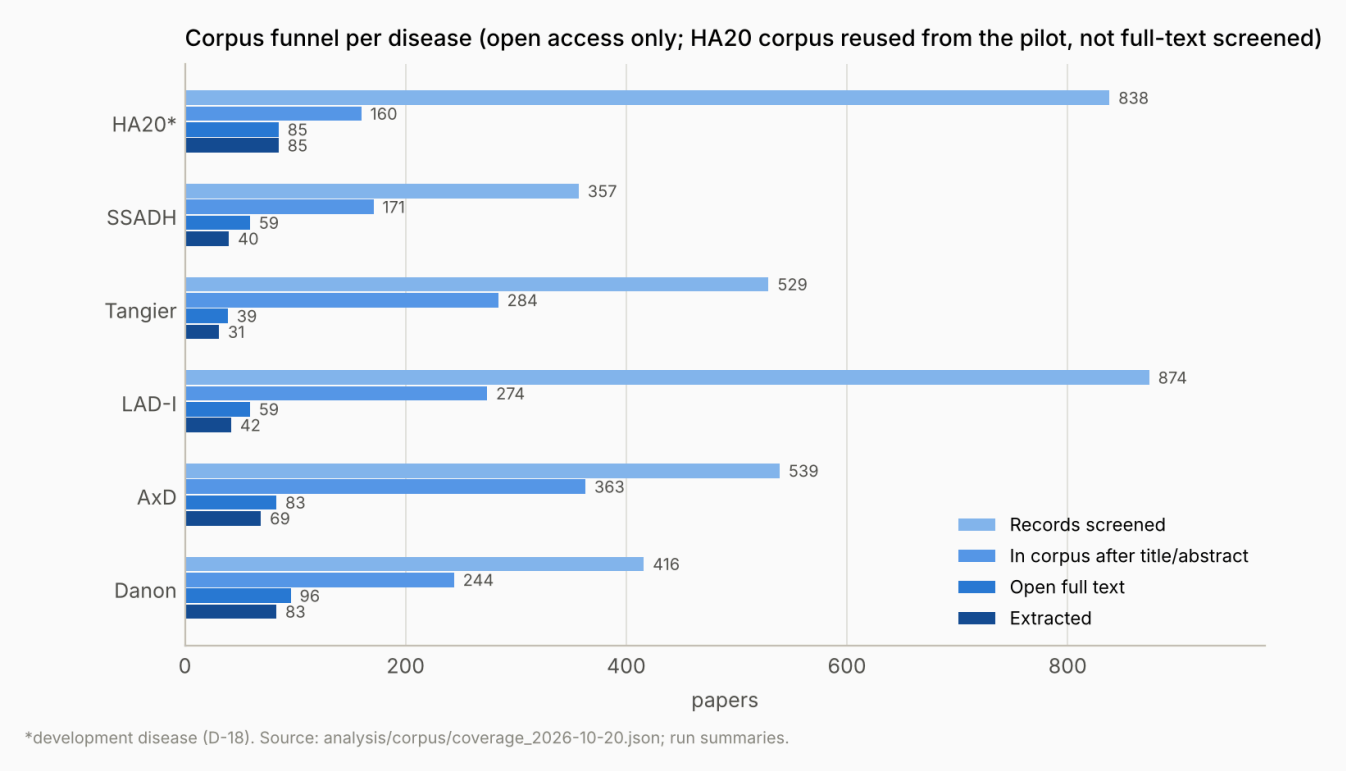


Figure 2. Corpus funnel per disease. Open-access-only access (D-32) is the binding constraint: 421 of 1,496 corpus papers (28%) have open full text, from 3,553 records screened.

- Open full text ranges from 14% (Tangier) to 53% (HA20) of the title/abstract corpus. Every disease cleared the D-08 floor of 15 eligible primary papers, so no substitute disease was needed.
- Paywalled papers are not a random subset. For HA20 they include the two largest sources of the Chen 2020 benchmark; for the other diseases the catalogue reviews' own source papers are often closed. The completeness denominators in §3.7 are small for that reason.
- Input problems found by the helpers during extraction (scrambled or image-only tables, missing or truncated supplements) are listed per disease in §3.12 and Supplementary Table S5. They bound what extraction could see and are not counted in any metric.

3.2 Extraction, provenance and the gatekeeper

Table 4. Extraction output and gatekeeper v4 verdicts. "Unsupported" counts unsupported and span-not-found verdicts; "not fully supported" adds partially supported. Intervals are exact. Source: `analysis/six_disease/run_summary_2026-10-20.json`.

Disease	Papers	Individual rows	Variant rows	Values with a quote	Span-check problems	Model claims judged	Unsupported	Not fully supported	Papers with any unsupported	Share held by worst 3 papers
HA20*	85	811	620	67,863	0	23,129	0.20% (46/23,129; exact 95% CI 0.15–0.27)	1.93% (446/23,129; exact 95% CI 1.75–2.11)	15 of 85	37%
SSADH	40	421	472	31,370	0	9,759	0.38% (37/9,759; exact 95% CI 0.27–0.52)	2.86% (279/9,759; exact 95% CI 2.54–3.21)	7 of 40	76%

Disease	Papers	Individual rows	Variant rows	Values with a quote	Span-check problems	Model claims judged	Unsupported	Not fully supported	Papers with any unsupported	Share held by worst 3 papers
Tangier	31	235	82	16,762	0	4,830	0.31% (15/4,830; exact 95% CI 0.17–0.51)	4.49% (217/4,830; exact 95% CI 3.93–5.12)	7 of 31	60%
LAD-I	42	508	411	36,280	0	10,864	0.43% (47/10,864; exact 95% CI 0.32–0.57)	3.47% (377/10,864; exact 95% CI 3.13–3.83)	12 of 42	66%
AxD	69	463	343	40,773	0	14,403	0.27% (39/14,403; exact 95% CI 0.19–0.37)	3.55% (512/14,403; exact 95% CI 3.26–3.87)	14 of 69	51%
Danon	83	510	305	48,682	0	17,963	0.61% (110/17,963; exact 95% CI 0.50–0.74)	3.07% (552/17,963; exact 95% CI 2.83–3.34)	23 of 83	55%
Five confirmatory	265	2,137	1,613	173,867	0	57,819	**0.43% (248/57,819; exact 95% CI 0.38–0.49)**	3.35% (1,937/57,819; exact 95% CI 3.20–3.50)	63 of 265	—
All six	350	2,948	2,233	241,730	0	80,948	**0.36% (294/80,948; exact 95% CI 0.32–0.41)**	2.94% (2,383/80,948; exact 95% CI 2.83–3.06)	78 of 350	—

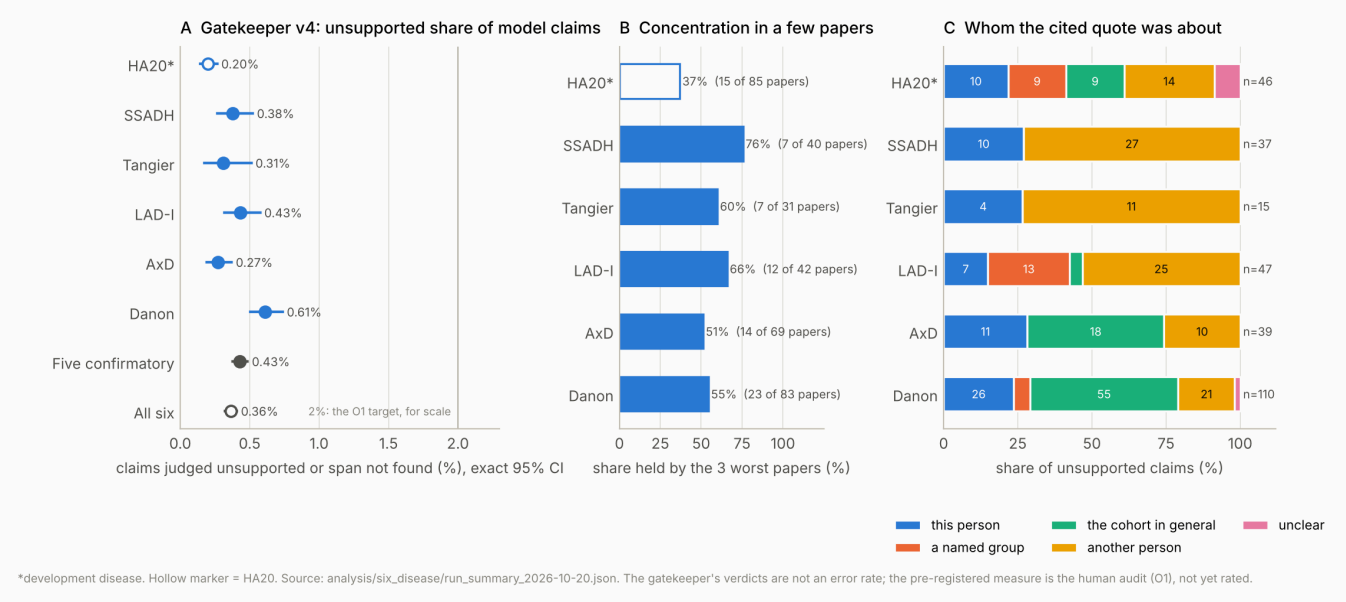


Figure 3. Gatekeeper v4 on the six runs. **A** Unsupported share with exact intervals, with the five-disease and six-disease pooled figures. **B** The share of each disease's unsupported claims held by its three worst papers. **C** Whom the cited quote was about, for unsupported claims only.

- **Provenance is complete.** Every one of the 241,730 values asserted across the six runs has a quote that the span check found verbatim in its paper (0 problems in every disease). Values without a quote are not emitted, so the fabricated-span count is zero by construction and by check.
- **The gatekeeper's unsupported share is small and similar across diseases:** 0.20% (HA20) to 0.61% (Danon). Pooled over the five confirmatory diseases it is 0.43% (248/57,819; exact 95% CI 0.38–0.49). **This is not an error rate.** It is what a second model read catches; the pilot showed it catching 1–2 of every 4–6 problems a person found (§3.6).
- **Unsupported claims cluster.** In five of six diseases the three worst papers hold half or more of that disease's unsupported claims (Figure 3B, Supplementary Figure S1). Danon's three worst papers hold 60 of its 110: variant and diagnosis fields given to eight family carriers when the paper names only the proband (37215540, 35 claims); "usually mild" applied to individuals (30413001, 15); exclusions resting on "typically showed no symptoms" (41560058, 10).
- **Why claims fail.** The commonest unsupported field is `variants.gene_symbol_as_stated` in five diseases (HA20 17, SSADH 16, Tangier 9, LAD-I 14, AxD 21, Danon 22 claims): a relative's or a cited patient's gene or variant quoted from a sentence about the proband. Extraction v5's rule (variant fields only from this person) holds for probands and slips for relatives the paper says carry "the same" variant. The second pattern is hedged wording ("suggested", "suspected", "consistent with", "typically") recorded as observed or excluded; extraction v6 has no hedge rule, a gap known and left in place at the freeze. The third is cohort-level statements applied to individuals (Figure 3C: "the cohort in general" dominates Danon's and AxD's unsupported claims). Partial verdicts are mostly ages at presentation recorded as onset or last evaluation.
- **Papers that contradict themselves.** The gatekeeper flagged about 30 papers whose table and text disagree (examples in Supplementary Table S5). Those claims were judged on the table and noted; they are collected for the human audit and for

adjudication.

3.3 Run-to-run stability

A random 10% of each disease's papers (ceil, seed 20261020; 38 papers) was re-extracted with the same frozen files and compared with the run of record. The gatekeeper was not run on the repeat.

Table 5. Agreement between the run of record and the repeat. Jaccard on sets per paper, pooled; checklist agreement on cells both runs have. Source: `analysis/six_disease/repeat_agreement_2026-10-20.json`.

Disease	Papers	Same individual count	Label Jaccard	Variants (strings)	Variants (folded keys)	Checklist agreement (cells)	Features, label-free	HPO status, label-free
HA20*	9	7 of 9	0.69	0.67	0.94	0.996 (1,300)	0.71	0.68
SSADH	4	4 of 4	1.00	1.00	1.00	1.000 (160)	0.43	0.71
Tangier	4	3 of 4	0.62	0.86	0.83	0.970 (200)	0.57	0.69
LAD-I	5	3 of 5	0.88	0.55	0.92	0.998 (440)	0.64	0.69
AxD	7	7 of 7	0.83	0.95	1.00	0.997 (1,340)	0.69	0.67
Danon	9	7 of 9	0.87	0.65	0.86	0.996 (680)	0.59	0.64

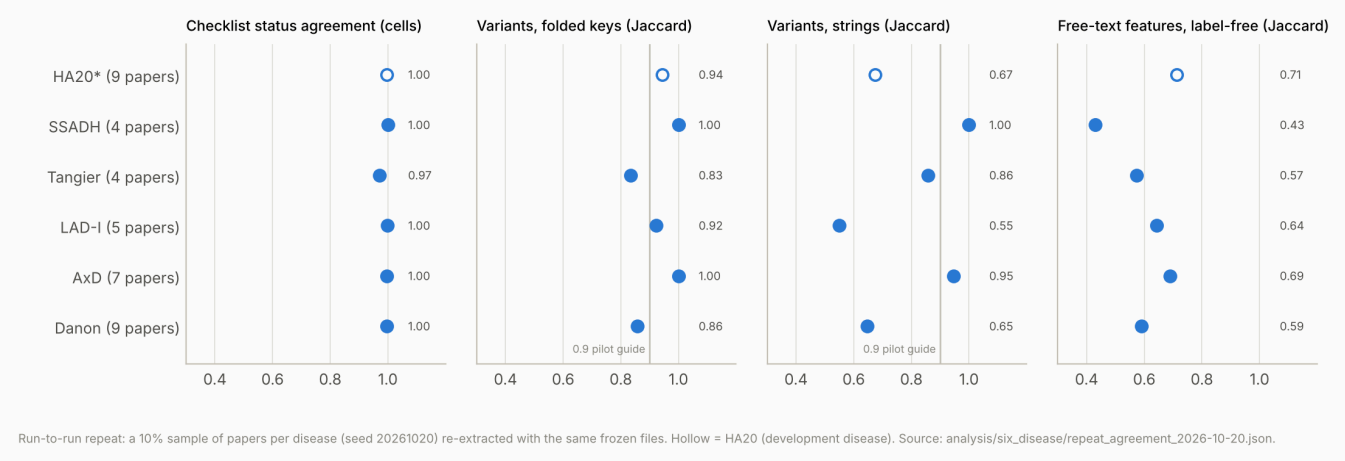


Figure 4. Run-to-run agreement by measure. Checklist statuses are stable everywhere; variant keys clear the pilot's 0.9 guide in four diseases; free-text features agree least because two readers phrase and split features differently.

- **Checklist statuses agree on 97–100% of cells** in every disease. Fixing the feature names removes the wording variance that free-text rows show (0.43–0.71 label-free).
- **Variant keys agree at 0.92–1.00 in four diseases** and fall to 0.83 (Tangier) and 0.86 (Danon), where one paper each is responsible and both runs recorded the same variants in different notation (a full-width asterisk " * " vs "*"; "genetic deletion of exons-4–10" vs "deletion of exons-4–10", which has no folded key).
- **Individual counts differ in 9 of 38 papers,** from scope calls on relatives and on patients cited from earlier reports, the same issue the pilot found in cohort papers. This is the measure to watch when the records are used for counting.

3.4 Records: reconciliation, normalization, binding and phenopackets

Table 6. Stage 4 records by disease. "Review-only" identities come only from a review's per-patient table (D-33). Normalization is counted over distinct cDNA descriptions; protein-only strings cannot be placed on the genome and are counted apart. Phenopackets keep only gatekeeper-supported claims; "claims dropped" is the count removed. Sources: `analysis/six_disease/stage4_summary_2026-10-20.json`, `reconcile_summary.json`, `phenopacket_build.json`, the phenopackets.

Disease	Rows → identities (merges)	Review-only identities	With target-gene variant	cDNA normalized	Feature rows bound to HPO	Phenopackets (with interpretation)	Terms observed / excluded	Observed–excluded conflicts	Claims dropped by gatekeeper	Sex	Schema-valid, errors
HA20*	811 → 769 (42)	282	558	96.7% of 195; 85 protein-only	91.6% of 5,079	769 (499)	3,678 / 1,018	93	207	348 F / 246 M / 175 unknown	769, 0
SSADH	421 → 363 (58)	43	243	86.5% of 147; 15 protein-only	89.8% of 1,683	363 (174)	1,249 / 299	10	151	90 F / 85 M / 188 unknown	363, 0
Tangier	235 → 235 (0)	140	65	90.0% of 31; 17 protein-only	90.7% of 1,051	235 (53)	745 / 219	13	117	88 F / 119 M / 28 unknown	235, 0

Disease	Rows → identities (merges)	Review-only identities	With target-gene variant	cDNA normalized	Feature rows bound to HPO	Phenopackets (with interpretation)	Terms observed / excluded	Observed–excluded conflicts	Claims dropped by gatekeeper	Sex	Schema-valid, errors
LAD-I	508 → 508 (0)	194	328	86.5% of 185; 7 protein-only	84.3% of 1,862	508 (299)	1,264 / 299	5	100	129 F / 146 M / 233 unknown	508, 0
AxD	463 → 456 (7)	175	318	78.9% of 116; 65 protein-only	91.3% of 3,316	456 (255)	2,441 / 529	40	221	149 F / 160 M / 147 unknown	456, 0
Danon	510 → 505 (5)	164	281	89.4% of 140; 25 protein-only	92.8% of 4,558	505 (212)	3,302 / 1,007	48	264	221 F / 243 M / 41 unknown	505, 0

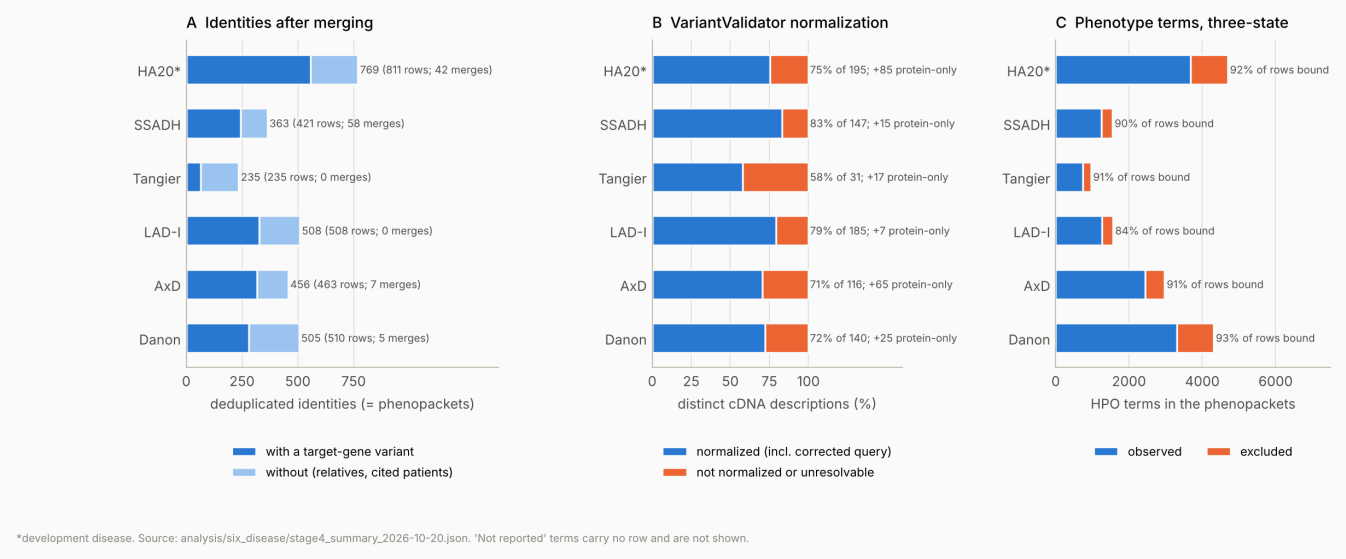


Figure 5. A Identities after merging, split by whether a target-gene variant is recorded. **B** VariantValidator normalization of distinct cDNA descriptions. **C** Observed and excluded HPO terms in the phenopackets.

- **2,836 identities from 2,948 rows; 2,836 phenopackets, all schema-valid; 1,492 carry a genomic interpretation.** Merge policy v4 made 112 merges in all; SSADH's 58 are all the duplicate study. Tangier and LAD-I have no citation merges, which is more likely a limit of the citation route than a fact about those literatures (the dossier helpers later proposed 19 same-person groups, §3.8).
- **Normalization** of cDNA descriptions ranges from 78.9% (AxD) to 96.7% (HA20). AxD papers often print cDNA without "c." ("739T→C"), and older papers use legacy numbering; 22 AxD descriptions were unresolvable. Protein-only strings (65 in AxD, 85 in HA20) can match ClinVar only at tier 2.
- **HPO binding** covers 84–93% of feature rows. The variant spectrum per disease (Supplementary Figure S2) matches expectation: missense dominates AxD (74%) and SSADH (51%); nonsense and frameshift dominate HA20 (63%) and Danon (52%).
- **Observed–excluded conflicts** (the same term observed and excluded for one identity, from different papers or rows) are few (5–93 per disease) and are kept visible in the records rather than resolved.
- **Sex** is unknown for 5–52% of identities by disease (SSADH highest), mostly people from review tables.
- Field coverage (Supplementary Figure S4) shows what the literature states: onset age is asserted for 3–50% of individuals and consanguinity for 0–30%, so the records carry "not reported" far more often than a value, as the three-state design requires.

3.5 O1: the unsupported-value audit (primary, confirmatory) — not yet rated

- **Drawn:** 250 claims from the 78,564 model claims gatekeeper v4 judged supported, stratified by disease in proportion to claim volume (HA20 72, SSADH 30, Tangier 15, LAD-I 33, AxD 44, Danon 56), seed 40250, on 2026-10-20, before any concordance was run. The draw records pool sizes, allocation, input hashes and the commit (`adjudication/o1_audit/sample_key.json`).
- **Built:** two blinded rating pages (one per rater) that show the claim, its quote and where the quote sits in the paper, never the gatekeeper's verdict; a collector that reports percent agreement and Cohen's κ, writes the tie-break sheet and applies rule C.
- **Not done:** no rating. The pages are private and unshared because the panel's availability (D-11) is unsettled, and Claude contacts no one. Until two raters finish, O1 has no value, and neither does the gatekeeper's accuracy against human labels (§7.1(b) of the plan).

3.6 What the pilot's human audits showed (HA20, before the freeze)

The pilot is the only place a person has rated claims. Two 50-claim audits by Dr. Lin, each on a different pipeline version, both missed the pilot's own 4% gate (Table 7). Each exposed one error class, and each class got a rule that is now frozen. The frozen pipeline has not been audited by a person.

Table 7. Pilot human audits (HA20; rated by Dr. Lin). Source: `analysis/poc/audit_metrics*.json`.

Audit	Pipeline	Claims rated	Unsupported (exact 95% CI)	Not fully supported	Error class found	Rule frozen in response
First	instructions v3, gatekeeper v1	50	3 of 50: 6.0% (1.3%–16.5%)	6 of 50	variant fields taken from a group statement or a literature-summary column, not this person	D-29: variant fields only from this person (v5)
Re-audit	instructions v5, gatekeeper v3 (20 papers)	48	3 of 48: 6.3% (1.3%–17.2%)	4 of 48	ages read by position from a stacked multi-patient cell; the rater could not verify them	D-31: no positional reading of stacked cells (v6, gatekeeper v4)
Re-audit, sensitivity	as above, without the 3 unverifiable claims	45	0 of 45: 0.0% (0.0%–7.9%)	1 of 45		

- **Variant fields were clean after D-29:** 0 of 20 unsupported in the re-audit, against 3 of 20 before.
- **The gatekeeper flagged few of the claims a person rejected:** v1 2 of 6, v3 1 of 4, with no false flags. Its own unsupported rate (0.036% for v1, 0.34% for v3) understated the human rate (6.0%, 6.3%) by an order of magnitude. That is why this report never presents Table 4's shares as error rates.
- **Deduplication:** merge policy v3 made 93 merges, of which a full hand check found 84 true, 6 false and 3 unclear; all 6 false merges came from the sex-and-onset route, which policy v4 dropped. Policy v4 had 0 false merges in the same set, but it was chosen after the check, so that is not an independent test; the registered 50-merge hand check of the six-disease run has not been done.
- **Completeness against Chen 2020 (pilot):** 9 of 9 restricted variants, 28 of 32 unrestricted.
- **Caveats carried forward:** the audits were small (n = 50, 48), rated by one person, and the second-look tabs showed the gatekeeper's reasons, so some labels there are not independent of it. HA20 is the development disease: every rule was tuned on it.

3.7 O2: concordance with ClinVar (primary, estimation); OMIM not assessed

Run on 2026-10-20 against the ClinVar release of 2026-10-12 (snapshot 2026-10-19), after the O1 sample was drawn. OMIM is not checked (no access, D-12). No threshold applies (D-03).

Table 8. Concordance with ClinVar by disease. Wilson 95% intervals. Gene (7.2.1) is near ceiling by design. HGVS recall counts ClinVar's own (paper, variant) citation pairs; overlap is not precision. Source: `analysis/six_disease/o2/o2_<d>.json`.

Disease	Gene matches disease gene	HGVS recall, tier 1	HGVS recall, tiers 1+2	Overlap, tier 1	Classification agreement (stated)	ClinVar condition names the disease	HPO annotation recall	Terms in ≥2 people matching an annotation	Corpus PMIDs cited by ClinVar
HA20*	99.8% (558/559; 95% CI 99.0–100.0)	26.9% (21/78; 95% CI 18.3–37.7)	26.9% (21/78; 95% CI 18.3–37.7)	37.9% (39/103; 95% CI 29.1–47.5)	82.8% (24/29; 95% CI 65.5–92.4)	47.4% (27/57; 95% CI 35.0–60.1)	92.9% (13/14; 95% CI 68.5–98.7)	6.1% (19/309; 95% CI 4.0–9.4)	20.0% (17/85; 95% CI 12.9–29.7)
SSADH	100.0% (243/243; 95% CI 98.4–100.0)	10.1% (16/158; 95% CI 6.3–15.8)	14.6% (23/158; 95% CI 9.9–20.9)	87.2% (68/78; 95% CI 78.0–92.9)	91.8% (45/49; 95% CI 80.8–96.8)	98.8% (83/84; 95% CI 93.6–99.8)	94.1% (32/34; 95% CI 80.9–98.4)	40.0% (52/130; 95% CI 32.0–48.6)	45.0% (18/40; 95% CI 30.7–60.2)
Tangier	98.5% (65/66; 95% CI 91.9–99.7)	16.7% (1/6; 95% CI 3.0–56.4)	50.0% (3/6; 95% CI 18.8–81.2)	56.2% (9/16; 95% CI 33.2–76.9)	80.0% (8/10; 95% CI 49.0–94.3)	37.5% (6/16; 95% CI 18.5–61.4)	70.8% (17/24; 95% CI 50.8–85.1)	15.3% (15/98; 95% CI 9.5–23.7)	16.1% (5/31; 95% CI 7.1–32.6)
LAD-I	100.0% (328/328; 95% CI 98.8–100.0)	23.3% (14/60; 95% CI 14.4–35.4)	31.7% (19/60; 95% CI 21.3–44.2)	34.9% (45/129; 95% CI 27.2–43.4)	70.0% (7/10; 95% CI 39.7–89.2)	90.4% (47/52; 95% CI 79.4–95.8)	70.6% (12/17; 95% CI 46.9–86.7)	14.4% (17/118; 95% CI 9.2–21.9)	16.7% (7/42; 95% CI 8.3–30.6)
AxD	99.7% (318/319; 95% CI 98.2–99.9)	27.5% (14/51; 95% CI 17.1–40.9)	66.7% (34/51; 95% CI 53.0–78.0)	73.1% (49/67; 95% CI 61.5–82.3)	58.6% (17/29; 95% CI 40.7–74.5)	85.6% (95/111; 95% CI 77.9–90.9)	84.2% (16/19; 95% CI 62.4–94.5)	13.5% (36/267; 95% CI 9.9–18.1)	39.1% (27/69; 95% CI 28.5–50.9)
Danon	97.9% (281/287; 95% CI 95.5–99.0)	35.0% (7/20; 95% CI 18.1–56.7)	50.0% (10/20; 95% CI 29.9–70.1)	47.9% (35/73; 95% CI 36.9–59.2)	78.9% (15/19; 95% CI 56.7–91.5)	89.6% (43/48; 95% CI 77.8–95.5)	87.1% (27/31; 95% CI 71.1–94.9)	15.9% (40/251; 95% CI 11.9–21.0)	18.1% (15/83; 95% CI 11.3–27.7)

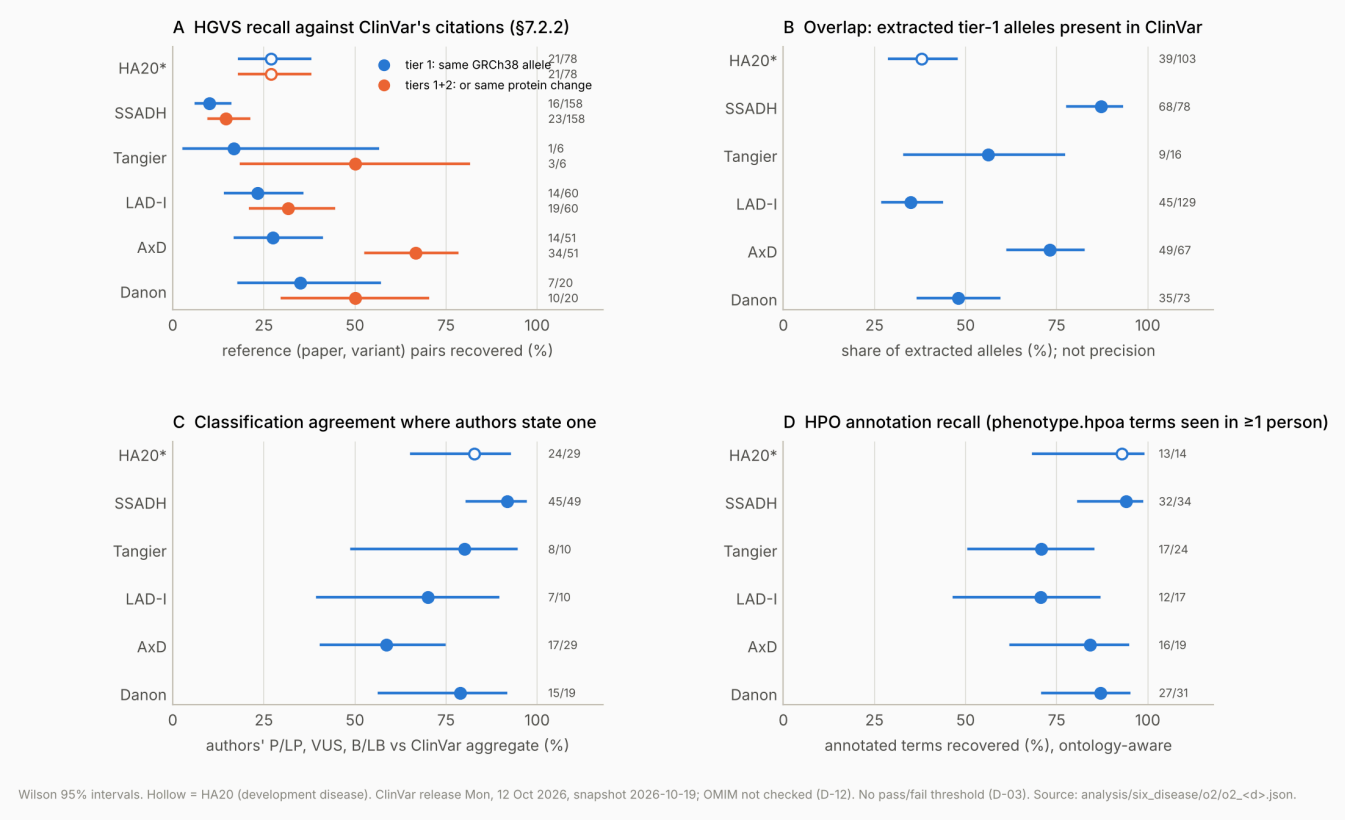


Figure 6. Concordance with ClinVar. **A** HGVS recall against ClinVar's citations. **B** Overlap of extracted tier-1 alleles with ClinVar. **C** Classification agreement where the authors state a class. **D** HPO annotation recall.

- **HGVS recall is low (10–35% at tier 1; 15–67% with protein-level matches) and the misses sit in single papers** (Supplementary Figure S7). Zhou 2016 (26642243) holds 52 of HA20's 57 misses: it is the founding paper, which ClinVar cites for many variants it does not describe. Van de Vijver 2012 (22134107) holds 37 of LAD-I's 41: a mutation-update review whose misses may be notation or normalization failures rather than scope. In SSADH, Akaboshi 2003 (14635103, whose table was scrambled in conversion) holds 68 and Brennenstuhl 2020 (33203024, variants tested in vitro with no patients) 53 of 135. Every miss counts as discordant in the primary figure; the registered triage of a seeded sample of 50 misses (`adjudication/o2_triage/`, seed 72650) will sort them into extraction error, normalization error, paper error, reference error and "citation does not describe the variant". For 28 of the 50, a script found no mention of the variant's position or residue in the paper text, which points at the last category or at a supplement we lack.
- **Overlap** (extracted tier-1 alleles present in ClinVar) is 35–87%. The low values (LAD-I 35%, HA20 38%) are where the gap candidates are (§3.8); the high value (SSADH 87%) is a gene whose literature ClinVar already covers well.
- **Classification agreement among stated values is 59–92%.** The disagreements run one way: authors assert P/LP where ClinVar holds VUS (Table 9). That is the directional expectation stated in advance (case-report authors assert causality; ClinVar's VUS and conflicting records limit concordance), not evidence of extraction error, which only O1 measures.
- **ClinVar's condition names the disease** for 38–99% of matched variants; the low values (Tangier 38%, HA20 47%) reflect records filed under broader or neighboring conditions.
- **Phenotype profile:** 71–94% of the HPO terms annotated to each disease are observed in at least one person; only 6–40% of terms seen in two or more people match an annotation. Annotations are known to be incomplete, so the second number mostly measures how much the literature reports beyond them.

Table 9. Classification: ClinVar aggregate (rows) against the authors' assertion. Source: the confusion matrices in `o2_<d>.json`.

Disease	ClinVar P/LP, authors P/LP	ClinVar VUS, authors P/LP	ClinVar P/LP, authors VUS	Authors state no class (any ClinVar class)	Agreeing pairs
HA20*	17	2	1	27	24
SSADH	38	3	1	34	45
Tangier	3	1	1	6	8
LAD-I	5	3	0	42	7
AxD	14	4	2	82	17
Danon	14	3	0	29	15

Without ClinVar-cited papers (D-19). Recall is undefined here, because every reference pair comes from a paper ClinVar cites. Overlap and classification agreement move little:

Disease	Overlap, tier 1	Classification agreement	Condition names disease
HA20*	33.3% (29/87; 95% CI 24.3–43.8)	85.7% (18/21; 95% CI 65.4–95.0)	48.9% (23/47; 95% CI 35.3–62.8)
SSADH	83.3% (40/48; 95% CI 70.4–91.3)	93.5% (29/31; 95% CI 79.3–98.2)	97.6% (41/42; 95% CI 87.7–99.6)
Tangier	57.1% (8/14; 95% CI 32.6–78.6)	85.7% (6/7; 95% CI 48.7–97.4)	33.3% (4/12; 95% CI 13.8–60.9)
LAD-I	40.0% (12/30; 95% CI 24.6–57.7)	62.5% (5/8; 95% CI 30.6–86.3)	100.0% (15/15; 95% CI 79.6–100.0)
AxD	69.4% (34/49; 95% CI 55.5–80.5)	66.7% (14/21; 95% CI 45.4–82.8)	87.2% (41/47; 95% CI 74.8–94.0)
Danon	41.5% (22/53; 95% CI 29.3–54.9)	80.0% (12/15; 95% CI 54.8–93.0)	90.0% (27/30; 95% CI 74.4–96.5)

Completeness against published catalogues (§7.2.7; secondary, descriptive). Only two diseases have a suitable open catalogue (Table 10, Supplementary Figure S5). The AxD catalogue was added after extraction (D-39), on the harness side only.

Table 10. Completeness. Restricted = catalogue variants whose source paper is in the corpus; unrestricted = all catalogue variants with a PMID. Source: `analysis/six_disease/completeness_pv1-<d>.json`.

Disease	Catalogue	Variants (patients)	Restricted denominator	Restricted recall, primary papers	Unrestricted recall	Restricted misses
HA20*	Chen 2020 Table S1 (D-30)	32 (89)	9	100.0% (9/9; 95% CI 70.1–100.0)	90.6% (29/32; 95% CI 75.8–96.8)	none
AxD	Grossi 2024 Suppl. Table 2 (D-39)	183 (553)	56	96.4% (54/56; 95% CI 87.9–99.0)	53.8% (91/169; 95% CI 46.3–61.2)	p.Met1_Met432del (36425948); p.Arg270_Ala272del (26478912)

- AxD's two restricted misses are notation, not absence: "R270-A272del" extracted with a hyphen where the catalogue has p.Arg270_Ala272del, and a whole-gene deletion extracted as "deletion (exons 1–9)". The matcher was not changed after seeing this.
- The restricted denominators are small (9 of 32, 56 of 183) because most catalogue source papers are not open access (D-32). Unrestricted recall (91% HA20, 54% AxD) therefore mostly measures access, not extraction.
- Patients per source paper: for AxD the run has at least as many carriers as the catalogue in all 43 shared source papers; for HA20 two papers have fewer (30022980: 1 of 4; 30526650: 2 of 3), candidates for a scope check.

3.8 O3: gap closure (primary, descriptive) — candidates only

A gap candidate is a tier-1 allele absent from the ClinVar snapshot whose identifying values (gene, strings, cDNA, protein, transcript, coordinates, person label) gatekeeper v4 judged supported in at least one row. OMIM allelic variants are not checked (D-12), so every candidate is a ClinVar gap only, and the OMIM half of the definition is unassessed.

Table 11. Gap candidates by disease. Protein check = the paper's stated protein against the one VariantValidator predicts from the normalized cDNA (a pre-triage flag). Reclassification candidates are in ClinVar only as VUS or conflicting; "strong" is a proposed rule (≥2 papers or ≥3 deduplicated people), exploratory. Source: `adjudication/o3_gaps/summary.json`.

Disease	Tier-1 alleles	Absent from ClinVar	Gap candidates	Absent but every row failed gatekeeper	Only from ClinVar-cited papers	Only from reviews	Protein check differs	Papers per candidate 1 / 2 / 3+	Authors' class	Reclassification (strong)
HA20*	103	64	**64**	0	6	0	8	43 / 14 / 7	43 P/LP, 4 VUS, 14 not stated	12 (6)
SSADH	78	10	**10**	0	2	0	1	6 / 4 / 0	6 P/LP, 0 VUS, 4 not stated	15 (4)
Tangier	16	7	**7**	0	1	0	1	7 / 0 / 0	4 P/LP, 0 VUS, 3 not stated	6 (0)
LAD-I	129	84	**82**	2	66	34	10	76 / 6 / 0	4 P/LP, 0 VUS, 78 not stated	7 (2)
AxD	67	18	**17**	1	2	1	3	15 / 2 / 0	8 P/LP, 3 VUS, 6 not stated	8 (2)
Danon	73	38	**35**	3	7	0	2	27 / 6 / 2	16 P/LP, 0 VUS, 19 not stated	5 (1)
Five confirmatory	363		**151**							41 (9)
All six	466	221	**215**	6						53 (15)

Stage (plan §7.3)	Count
1 Candidates	215

Stage (plan §7.3)	Count
2 Adjudicated by the panel	0
3 Approved	0
4 Drafted (pending adjudication)	0
5 Signed off	0
6 Submitted	0
7 Accepted	0

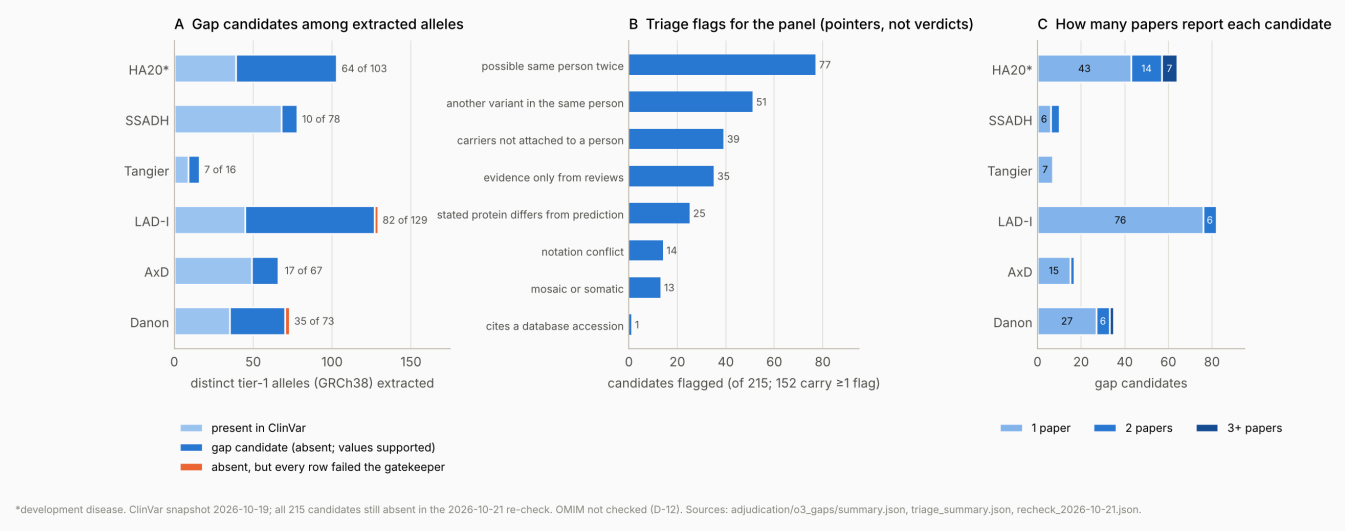


Figure 7. Gap candidates. **A** Extracted tier-1 alleles split by presence in ClinVar. **B** Keyword triage flags drawn from the evidence dossiers' concerns, as pointers for the panel. **C** How many papers report each candidate.

- **215 candidates** (151 outside HA20), from 466 tier-1 alleles. All 215 were still absent by GRCh38 allele in the next ClinVar release (variant summary of 19 Oct 2026, snapshot 2026-10-21). A spot check found that HA20's c.271A>T (p.Lys91Ter) and c.559C>T (p.Gln187Ter) have no record while their neighbors do.
- **LAD-I's 82 need the most care.** 66 come only from papers ClinVar cites for *ITGB2* and 34 only from reviews, most from van de Vijver 2012, a mutation update. A variant that ClinVar cites a paper for but that is absent by allele is as likely a numbering or normalization slip as a true gap; the protein check flags 10 of them.
- **Evidence dossiers** were written for all 215 candidates by helpers reading only the extracted records and the papers (never ClinVar or OMIM): case count and phenotype match, segregation, de novo status with parental testing, functional evidence, in-trans findings and variant type, each line with a PMID and a verbatim quote or "not reported", and no proposed classification. All 4,725 quotes were checked verbatim; 0 problems. Population frequency (PM2) is left to the panel because gnomAD is not reachable here.
- **Triage flags** (Figure 7B): 152 of 215 candidates carry at least one keyword flag from the dossiers' concerns. The most common, "possible same person twice" (77), matters for case counts, not for whether the variant exists. The dossier helpers also named candidates that may not be gaps at all: a coordinate read literally from "c.610-612A>G (IVS3-2A>G)" (G-ssadh-003); c.54G>T called Gly18Val where c.53 is probably meant (G-axd-017); a paper that cites an SCV accession not yet in the release (G-lad1-077); two LAMP2B-numbered variants mapped to intronic positions on the MANE transcript (G-danon-002).
- **Proposed same-person groups:** 19 groups across 4 diseases (HA20 12, Danon 5, AxD 1, LAD-I 1) of identities the dossier helpers judged to be one person re-cited by a later paper. If all were confirmed, 3.8% of HA20 identities would collapse and 0.2%–1.2% elsewhere; a lower bound, because only the candidates' papers were examined. Nothing is merged: changing the records after the freeze is a deviation, and each group needs a person.
- **Reclassification candidates (exploratory):** 53 supported tier-1 variants are in ClinVar only as VUS or conflicting; 15 meet the proposed "strong" rule (Supplementary Table S9).
- **Submission records** are prepared as format-neutral pending records per panel-approved variant (`submissions/make_records.py`); the ClinVar collection method is *curation* (D-13). None exists yet because nothing is approved.

3.9 O4: ascertainment bias and memorization (secondary)

Memorization probe (exploratory, §7.4.3). 180 papers (30 per disease) stratified by first publication date against Claude Opus's documented cutoff (June 2026): 10 after, 51 recent (2024-07 to 2026-06), 119 older. Two black-box probes run identically on both models; every Claude helper transcript was checked for forbidden tool calls (none). Scored against the full-text extraction, never ClinVar.

Table 12. Memorization probe by stratum and model. Closed book = recall from PMID, title and year; completion = continuing a 50-word opening. One item (Tangier 33346003) was excluded because its opening was publisher boilerplate. Sources: `analysis/bias/memorization/scores_summary.json`, `openweight_scores_summary.json`.

Stratum	Papers	Model	Closed book: specific / guess / none	Papers with a variant hit	Variants given (hits)	Phenotype terms given, mapped, hits	Completion: mean ROUGE-L F1	Mean 5-gram share	Longest run (words)	Runs ≥ 8 words
after cutoff	10	Claude	0 / 6 / 4	0	0 (0)	24, 20, 12	0.158	0.0000	4	0
after cutoff	10	DeepSeek	0 / 3 / 7	0	0 (0)	9, 1, 1	0.137	0.0000	4	0
recent	51	Claude	0 / 44 / 7	2	9 (2)	163, 132, 68	0.190	0.0042	7	0
recent	51	DeepSeek	0 / 11 / 40	1	4 (1)	26, 2, 1	0.154	0.0017	6	0
older	119	Claude	4 / 95 / 20	2	3 (2)	306, 248, 153	0.190	0.0061	11	3
older	119	DeepSeek	0 / 27 / 92	2	9 (2)	68, 31, 23	0.148	0.0021	7	0

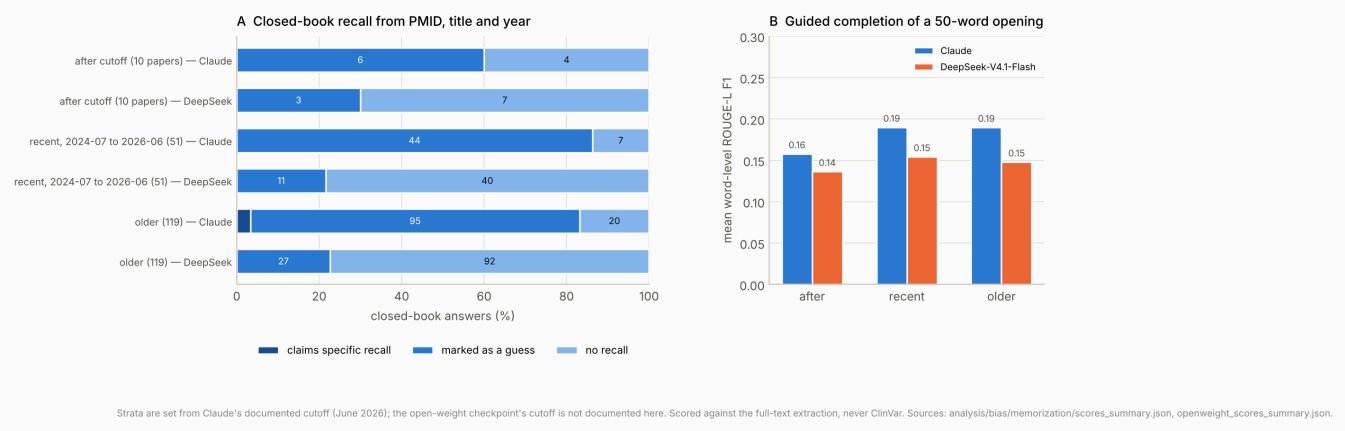


Figure 9. Memorization probe on both models. Neither reproduces the papers' variants or text; the differences between strata are small and in no consistent direction.

- Claude claimed specific recall for 4 of 180 papers, all older, and in each case for phenotypes only; 4 papers had a variant hit, all labelled guesses and all recurrent variants (e.g. LAMP2 p.Arg293Ter). DeepSeek claimed specific recall for none and answered "none" for 139 papers. The three completions with a run of 8+ matching words are all formulaic ("written informed consent was obtained from all patients").
- The after-cutoff stratum has 10 papers, so the before/after comparison has little power; and black-box probes can miss memorization that changes extraction without surfacing as recall. The exploratory split of audit and concordance results by memorization has nothing to split on.

Frequency deviation (§7.4.2): not started. 4 open comparators are fetched and a listing sheet exists for each (AxD: gagliardi2024_axd_ukb; Danon: khedro2025_danon_prospective; HA20: he2026_ha20_multicenter; SSADH: dibacco2019_ssadh_natural_history); 3 candidate comparators are paywalled and left out (D-32); LAD-I and Tangier have no comparator and are out of the bias arm (D-35). Step 1 requires a person to list each comparator's HPO-mappable features and cohort frequencies before any literature frequency is computed, and the script that then runs the deviation (with a 2,000-resample publication-cluster bootstrap, seed 7425) is written and tested on synthetic input only.

3.10 O5: efficiency and the open-weight comparison (secondary)

Expert-minutes per accepted record: not measured. The adjudication page logs active minutes per record, but adjudication has not started.

Claude vs DeepSeek-V4.1-Flash (D-10). The open-weight model (2026-09-10 checkpoint, OpenRouter, provider pinned to DeepInfra fp8) ran the same 85 HA20 and 40 SSADH papers with the same launch prompt, frozen instruction files and schema, through an agent loop with file tools (`pipeline/extract/openweight/agent.py`). Outputs went through the same span check, reconciliation, normalization and O2 harness, and the same gatekeeper v4 judged both runs' claims.

Table 13. Claude vs open-weight model. Values are Claude / open-weight. Claude's cost is an estimate from helper tokens at the six-disease run's rate; the open-weight cost is what the host reported for final attempts (failed attempts, 13 HA20 and 1 SSADH papers, add $0.55 logged plus an unknown amount for 10 attempts that stopped before logging). Sources: `analysis/openweight/summary_<d>.json`, `gatekeeper_compare.json`, `o2/`.

Disease	Papers	Values asserted	Span-not-found rate	Values with no quote	Judged unsupported (gatekeeper v4)	...among claims with a quote	HGVS recall, tiers 1+2	Variant keys shared/union	Checklist agreement (shared cells)	Feature rows	Cost per paper, median	Wall-clock per paper, median (min)
HA20*	85	67,863 / 55,317	0.00% / **3.86%**	0 / 2,003	**0.20%** / **8.64%**	0.20% / 2.28%	26.9% (21/78; 95% CI 18.3–37.7) / 19.2% (15/78; 95% CI 12.0–29.3)	261/294 (88.8%)	95.9% of 10,720	5,079 / 2,753	$1.08 / $0.07	2.9 / 20.9
SSADH	40	31,370 / 33,521	0.00% / **2.98%**	0 / 991	**0.38%** / **6.14%**	0.38% / 2.03%	14.6% (23/158; 95% CI 9.9–20.9) / 15.2% (24/158; 95% CI 10.4–21.6)	157/166 (94.6%)	97.6% of 6,716	1,683 / 1,439	$1.19 / $0.07	2.9 / 19.2

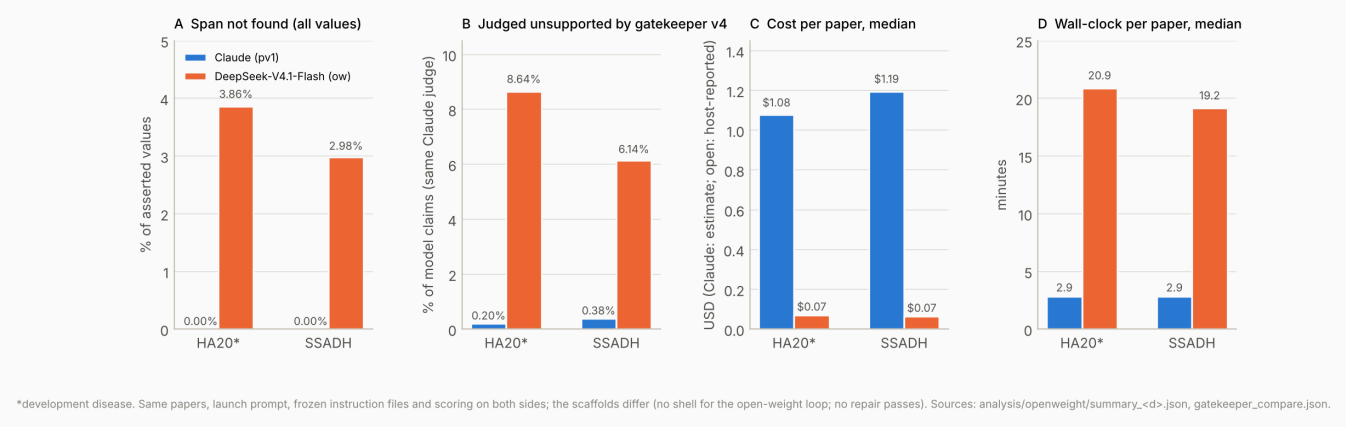


Figure 8. Claude vs the open-weight model on the registered metrics measured so far.

- **Variants:** the two models found largely the same target-gene variants (88.8% and 94.6% of the union shared) and their HGVS recall against ClinVar overlaps within the intervals.
- **Provenance:** Claude left no value without a quote; the open-weight model left 3.6% (HA20) and 3.0% (SSADH) without one, and 45 and 6 cells broke the schema shape (tolerated by the readers after checking pv1 outputs were byte-identical).
- **Support, as the same Claude gatekeeper reads it:** 8.6% and 6.1% of the open-weight model's claims unsupported against 0.20% and 0.38% for Claude. Most of the gap is claims with no quote (`is_proband = false`, checklist statuses), but claims with a quote were also judged unsupported 11× (HA20) and 5× (SSADH) as often. Two cautions: the judge is Claude, which may not be neutral between a Claude and a non-Claude extraction, and helpers did not all treat empty quotes the same way (236 HA20 and 144 SSADH empty-quote claims were judged supported against the row's quote). The blinded human audit of equal samples from each model, the registered measure, waits on D-11.
- **Cost and time:** per paper the open-weight model cost about 1/15 of Claude's estimate and took 7× as long (one provider's throughput, with waits after dropped connections: 36 and 17 retried calls in final attempts, ~50 per 125 papers).
- **Scaffold differences, as the plan requires them stated:** the open-weight loop had no shell (Claude's helpers used one for small checks); it had no repair passes for empty quotes (pv1 had a handful); three harness fixes landed during the run (retries, a parse check on JSON writes, a resume request after a cut-off reply, never needed), approved by Dr. Lin and logged.
- **Scope:** two diseases, one open-weight model, one provider; not general findings.

3.11 Exploratory analyses (§7.6)

Genotype–phenotype maps with GPSEA. Fisher exact tests with Benjamini–Hochberg correction at 0.05, in a separate uv project (GPSEA 0.10.0; pandas < 3). Variant effects come from each phenopacket's own normalized HGVS (Ensembl VEP is not reachable here). Only individuals with a variant placed on GRCh38 and at least one phenotype enter. Zygosity for untyped calls is an analysis assumption only (dominant: heterozygous; Danon males hemizygous; recessive: two distinct untyped calls taken as compound heterozygous); the records keep "not reported". A feature not reported is not counted as excluded.

Table 14. GPSEA tests. Source: `analysis/exploratory/gpsea/results/summary.json`.

Disease	Comparison	Individuals	Terms tested	Significant (BH 0.05)	Significant terms (group shares; p_BH)
HA20*	truncating vs missense	217	11	0	—
SSADH	biallelic: truncating/truncating vs truncating/other vs other/other	77	9	0	—
Tangier	biallelic: truncating/truncating vs truncating/other vs other/other	16	0	0	—

Disease	Comparison	Individuals	Terms tested	Significant (BH 0.05)	Significant terms (group shares; p_BH)
LAD-I	biallelic: truncating/truncating vs truncating/other vs other/other	38	3	0	—
AxD	hotspot R79/R88/R239 vs other missense	83	14	0	—
Danon	truncating vs missense	130	9	2	Arrhythmia (truncating 57/59, missense 4/7; p_BH 0.035); Ventricular arrhythmia (truncating 45/47, missense 3/6; p_BH 0.035)
Danon	sex (males hemizygous, females heterozygous)	130	10	3	Proximal muscle weakness (FEMALE 1/28, MALE 11/25; p_BH 0.005); Left ventricular hypertrophy (FEMALE 25/33, MALE 41/41; p_BH 0.005); Hypertrophic cardiomyopathy (FEMALE 19/24, MALE 34/34; p_BH 0.031)

- **Danon reproduces the published sex difference** (Supplementary Figure S6): proximal muscle weakness 11/25 males vs 1/28 females; left ventricular hypertrophy 41/41 vs 25/33; hypertrophic cardiomyopathy 34/34 vs 19/24. The papers overlap with the published series, so this is a reproduction check, not independent replication. The truncating-vs-missense arrhythmia signal rests on 6–8 missense carriers and is a lead only.
- **HA20, AxD, SSADH, LAD-I:** no term survives correction; Tangier has too few placeable, phased genotypes to test.

Tangier: share of ClinVar *ABCA1* records matched by the literature. 16 of 2,006 single-gene GRCh38 records (0.8%) and 7 of 85 P/LP records (8.2%) match variants extracted from the 31-paper Tangier corpus; 19 of 35 extracted variants are not in ClinVar. A lower bound: the corpus is open access only, and most *ABCA1* records come from testing unrelated to Tangier disease.

LAD-I deduplication accuracy against the duplicate cases in Almarza Novoa 2018 is not computable: the review's open supplement flags only one overlap (10 cases in a 1987 paper outside the corpus) and names no case pairs.

3.12 Results by disease

Each subsection gives the disease's corpus, records, variants, gatekeeper reading, stability, concordance, gap candidates, what the helpers flagged in its inputs, and any disease-specific finding. HA20 comes first because it is the development disease; its figures are secondary.

A20 haploinsufficiency (HA20, *TNFAIP3*) — development disease

- **Corpus:** 838 records screened; 160 in corpus; 85 with open full text (53.1%); 85 papers extracted, 77 with individual rows. ClinVar (12 Oct 2026): 764 single-gene records, 99 P/LP, 375 VUS.
- **Records:** 811 rows → 769 identities (42 merges; 282 review-only); 558 with a *TNFAIP3* variant; 769 phenopackets, 499 with a genomic interpretation; sex 348 F / 246 M / 175 unknown. Observed / excluded terms 3,678 / 1,018; most-observed: Oral ulcer 43%; Recurrent fever 27%; Inflammatory abnormality of the skin 22%; Genital ulcers 21%; Arthralgia/arthritis 17%.
- **Variants:** 280 distinct descriptions; cDNA normalized 96.7%; 523 interpretations in the phenopackets (frameshift 33%, nonsense 30%, missense 21%).
- **Gatekeeper:** 0.20% (46/23,129; exact 95% CI 0.15–0.27) unsupported; 15 of 85 papers; top field `variants.gene_symbol_as_stated` (17).
- **Repeat (9 papers):** checklist 0.996; variant keys 0.94; same individual count in 7 of 9.
- **ClinVar:** HGVS recall tier 1 26.9% (21/78; 95% CI 18.3–37.7), tiers 1+2 26.9% (21/78; 95% CI 18.3–37.7); overlap 37.9% (39/103; 95% CI 29.1–47.5); classification agreement 82.8% (24/29; 95% CI 65.5–92.4); HPO annotation recall 92.9% (13/14; 95% CI 68.5–98.7). O2 misses: 57 in 6 papers, led by 26642243 (52).
- **Gap candidates:** 64 of 103 tier-1 alleles (43 in one paper, 14 in two, 7 in three or more); 6 only from ClinVar-cited papers; protein check differs for 8; 12 reclassification candidates (6 strong).
- **Development disease (D-18).** Every rule in extract_v6 and gatekeeper_v4 was tuned on HA20's pilot runs, and 92 of its 160 corpus papers were seeded from ClinVar citations (D-19). Treat its figures as an upper bound on the pipeline.
- **Completeness vs Chen 2020:** 9 of 9 restricted variants, 29 of 32 unrestricted; most of the catalogue's source papers are paywalled.
- **GPSEA:** truncating vs missense, 217 individuals, 11 terms, none significant.
- **Same-person groups:** 12 proposed groups; up to 3.8% of identities would collapse if confirmed, the highest of any disease, because HA20 reviews re-cite the same families.

SSADH deficiency (SSADH, *ALDH5A1*)

- **Corpus:** 357 records screened; 171 in corpus; 59 with open full text (34.5%); 40 papers extracted, 40 with individual rows. ClinVar (12 Oct 2026): 914 single-gene records, 152 P/LP, 333 VUS.
- **Records:** 421 rows → 363 identities (58 merges; 43 review-only); 243 with a *ALDH5A1* variant; 363 phenopackets, 174 with a genomic interpretation; sex 90 F / 85 M / 188 unknown. Observed / excluded terms 1,249 / 299; most-observed: Global developmental delay 25%; Seizure 20%; Hypotonia 18%; Elevated urinary 4-hydroxybutyric acid 13%; Abnormal basal ganglia MRI signal intensity 12%.

- **Variants:** 162 distinct descriptions; cDNA normalized 86.5%; 286 interpretations in the phenopackets (missense 51%, nonsense 15%, not placed 13%).
- **Gatekeeper:** 0.38% (37/9,759; exact 95% CI 0.27–0.52) unsupported; 7 of 40 papers; top field `variants.gene_symbol_as_stated` (16).
- **Repeat (4 papers):** checklist 1.000; variant keys 1.00; same individual count in 4 of 4.
- **ClinVar:** HGVS recall tier 1 10.1% (16/158; 95% CI 6.3–15.8), tiers 1+2 14.6% (23/158; 95% CI 9.9–20.9); overlap 87.2% (68/78; 95% CI 78.0–92.9); classification agreement 91.8% (45/49; 95% CI 80.8–96.8); HPO annotation recall 94.1% (32/34; 95% CI 80.9–98.4). O2 misses: 135 in 9 papers, led by 14635103 (68).
- **Gap candidates:** 10 of 78 tier-1 alleles (6 in one paper, 4 in two, 0 in three or more); 2 only from ClinVar-cited papers; protein check differs for 1; 15 reclassification candidates (4 strong).
- **Input problems flagged by the helpers:** 14635103: Table 2 scrambled in conversion; about 12 of 54 families extracted; 38499966: table garbled; supplements S1/S2 missing; 37503297 and 37962671 are the same study (preprint and journal version); their patients are merged (decision 2026-10-20).
- **Duplicate study:** 37503297 (Research Square preprint) and 37962671 (journal version, whose open copy is the preprint text) are both in the corpus; their patients are merged (58 merges) with the evidence recorded. The papers themselves disagree on G176R vs G176E for one variant.
- **O2 misses are two papers:** Akaboshi 2003 (68 misses; its Table 2 was scrambled in conversion, so about 12 of 54 families were extracted) and Brennenstuhl 2020 (53; variants tested in vitro, no patients). Both are scope or input problems rather than reading errors, pending triage.
- **Overlap with ClinVar is the highest of the six (87.2% (68/78; 95% CI 78.0–92.9)),** and ClinVar's condition names SSADH deficiency for 98.8% (83/84; 95% CI 93.6–99.8) of matched variants: a well-covered gene. Only 10 gap candidates.
- **GPSEA:** biallelic truncating vs other, 77 individuals, 9 terms, none significant.

Tangier disease (Tangier, *ABCA1*)

- **Corpus:** 529 records screened; 284 in corpus; 39 with open full text (13.7%); 31 papers extracted, 31 with individual rows. ClinVar (12 Oct 2026): 2,006 single-gene records, 85 P/LP, 705 VUS.
- **Records:** 235 rows → 235 identities (0 merges; 140 review-only); 65 with a *ABCA1* variant; 235 phenopackets, 53 with a genomic interpretation; sex 88 F / 119 M / 28 unknown. Observed / excluded terms 745 / 219; most-observed: Decreased circulating HDL-C concentration 21%; Angina pectoris 14%; Coronary artery atherosclerosis 13%; Abnormality of the vasculature 13%; Decreased circulating apolipoprotein A-I concentration 11%.
- **Variants:** 48 distinct descriptions; cDNA normalized 90.0%; 65 interpretations in the phenopackets (missense 43%, not placed 25%, nonsense 20%).
- **Gatekeeper:** 0.31% (15/4,830; exact 95% CI 0.17–0.51) unsupported; 7 of 31 papers; top field `variants.gene_symbol_as_stated` (9).
- **Repeat (4 papers):** checklist 0.970; variant keys 0.83; same individual count in 3 of 4.
- **ClinVar:** HGVS recall tier 1 16.7% (1/6; 95% CI 3.0–56.4), tiers 1+2 50.0% (3/6; 95% CI 18.8–81.2); overlap 56.2% (9/16; 95% CI 33.2–76.9); classification agreement 80.0% (8/10; 95% CI 49.0–94.3); HPO annotation recall 70.8% (17/24; 95% CI 50.8–85.1). O2 misses: 3 in 3 papers, led by 19133158 (1).
- **Gap candidates:** 7 of 16 tier-1 alleles (7 in one paper, 0 in two, 0 in three or more); 1 only from ClinVar-cited papers; protein check differs for 1; 6 reclassification candidates (0 strong).
- **Input problems flagged by the helpers:** 26713019 and 27011649: tables published as images.
- **Smallest corpus and the lowest open-access share (13.7%).** Many Tangier papers are old and their tables are images (26713019, 27011649).
- **Only 65 of 235 identities carry an *ABCA1* variant:** 140 identities come from review tables, and many older clinical reports give no genotype. Papers on heterozygous carriers with familial HDL deficiency were kept as "unclear" at screening for the full text to decide.
- **ClinVar *ABCA1* is dominated by testing unrelated to Tangier disease:** 16 of 2,006 records match the literature (exploratory, §3.11).
- **No comparator** for the bias arm and no suitable open catalogue; GPSEA could not run (too few phased genotypes).

Leukocyte adhesion deficiency I (LAD-I, *ITGB2*)

- **Corpus:** 874 records screened; 274 in corpus; 59 with open full text (21.5%); 42 papers extracted, 42 with individual rows. ClinVar (12 Oct 2026): 949 single-gene records, 87 P/LP, 345 VUS.
- **Records:** 508 rows → 508 identities (0 merges; 194 review-only); 328 with a *ITGB2* variant; 508 phenopackets, 299 with a genomic interpretation; sex 129 F / 146 M / 233 unknown. Observed / excluded terms 1,264 / 299; most-observed: Increased total leukocyte count 32%; Neonatal omphalitis 12%; Pyoderma gangrenosum 11%; Reduced cell surface marker level 11%; Sepsis 10%.
- **Variants:** 192 distinct descriptions; cDNA normalized 86.5%; 361 interpretations in the phenopackets (missense 41%, nonsense 16%, frameshift 15%).
- **Gatekeeper:** 0.43% (47/10,864; exact 95% CI 0.32–0.57) unsupported; 12 of 42 papers; top field `variants.gene_symbol_as_stated` (14).

- **Repeat (5 papers):** checklist 0.998; variant keys 0.92; same individual count in 3 of 5.
- **ClinVar:** HGVS recall tier 1 23.3% (14/60; 95% CI 14.4–35.4), tiers 1+2 31.7% (19/60; 95% CI 21.3–44.2); overlap 34.9% (45/129; 95% CI 27.2–43.4); classification agreement 70.0% (7/10; 95% CI 39.7–89.2); HPO annotation recall 70.6% (12/17; 95% CI 46.9–86.7). O2 misses: 41 in 4 papers, led by 22134107 (37).
- **Gap candidates:** 82 of 129 tier-1 alleles (76 in one paper, 6 in two, 0 in three or more); 66 only from ClinVar-cited papers; protein check differs for 10; 7 reclassification candidates (2 strong).
- **Input problems flagged by the helpers:** 33391282: supplement Table S1 cut at 60,000 characters, so P75–P127 have no clinical rows.
- **The gap list is dominated by one review:** van de Vijver 2012 (22134107), a mutation update whose people are database accession numbers. 66 of the 82 candidates come only from ClinVar-cited papers and 34 only from reviews; the protein check flags 10. Many may be numbering or normalization slips rather than gaps. The same paper holds 37 of LAD-I's 41 O2 misses.
- **Input problem:** 33391282's Table S1 was cut at 60,000 characters, so P75–P127 have no clinical rows; 30 of its patients were reported before without names, which the merge route cannot join.
- **Gene-name spelling:** "ITGβ2", "ITGß2" and "CD18" cover 148 variant rows; the matcher accepts them.
- **GPSEA:** biallelic truncating vs other, 38 individuals, 3 terms, none significant. No comparator for the bias arm; the deduplication benchmark is not computable.

Alexander disease (AxD, *GFAP*)

- **Corpus:** 539 records screened; 363 in corpus; 83 with open full text (22.9%); 69 papers extracted, 69 with individual rows. ClinVar (12 Oct 2026): 627 single-gene records, 76 P/LP, 262 VUS.
- **Records:** 463 rows → 456 identities (7 merges; 175 review-only); 318 with a *GFAP* variant; 456 phenopackets, 255 with a genomic interpretation; sex 149 F / 160 M / 147 unknown. Observed / excluded terms 2,441 / 529; most-observed: Abnormal cerebral white matter morphology 19%; Seizure 16%; Dysarthria 14%; Gait disturbance 14%; Atrophy/Degeneration affecting the brainstem 12%.
- **Variants:** 181 distinct descriptions; cDNA normalized 78.9%; 261 interpretations in the phenopackets (missense 74%, not placed 16%, splice / intronic 3%).
- **Gatekeeper:** 0.27% (39/14,403; exact 95% CI 0.19–0.37) unsupported; 14 of 69 papers; top field `variants.gene_symbol_as_stated` (21).
- **Repeat (7 papers):** checklist 0.997; variant keys 1.00; same individual count in 7 of 7.
- **ClinVar:** HGVS recall tier 1 27.5% (14/51; 95% CI 17.1–40.9), tiers 1+2 66.7% (34/51; 95% CI 53.0–78.0); overlap 73.1% (49/67; 95% CI 61.5–82.3); classification agreement 58.6% (17/29; 95% CI 40.7–74.5); HPO annotation recall 84.2% (16/19; 95% CI 62.4–94.5). O2 misses: 17 in 10 papers, led by 28882119 (5).
- **Gap candidates:** 17 of 67 tier-1 alleles (15 in one paper, 2 in two, 0 in three or more); 2 only from ClinVar-cited papers; protein check differs for 3; 8 reclassification candidates (2 strong).
- **Input problems flagged by the helpers:** 34046764: Table 1 caption only; 32126152: Table S1 missing; 39420046 (Grossi 2024): supplement arrived after extraction (D-39).
- **Completeness vs Grossi 2024 (D-39):** 54 of 56 restricted variants (protein level 52/56, cDNA level 36/56); the two misses are notation. The catalogue's genomic column matched nothing as printed (30 off by one, 16 on the coding strand), so it was not used.
- **Normalization is the lowest (78.9%):** AxD papers print cDNA without "c." and use legacy numbering; 65 strings are protein-only, so 43 interpretations are unplaced and can match ClinVar only at tier 2 (tiers 1+2 recall 66.7% (34/51; 95% CI 53.0–78.0) vs tier 1 27.5% (14/51; 95% CI 17.1–40.9)).
- **Missense dominates (74% of interpretations)**, as expected for a gain-of-function leukodystrophy; GPSEA found no difference between the R79/R88/R239 hotspots and other missense (83 individuals, 14 terms).
- **Comparators:** Gagliardi 2025 (UK Biobank, genotype-first) is fetched and awaits a person; Mura 2021 is paywalled.

Danon disease (Danon, *LAMP2*)

- **Corpus:** 416 records screened; 244 in corpus; 96 with open full text (39.3%); 83 papers extracted, 83 with individual rows. ClinVar (12 Oct 2026): 986 single-gene records, 154 P/LP, 292 VUS.
- **Records:** 510 rows → 505 identities (5 merges; 164 review-only); 281 with a *LAMP2* variant; 505 phenopackets, 212 with a genomic interpretation; sex 221 F / 243 M / 41 unknown. Observed / excluded terms 3,302 / 1,007; most-observed: Left ventricular hypertrophy 39%; Hypertrophic cardiomyopathy 28%; Congestive heart failure 20%; Myocardial late gadolinium enhancement 17%; Ventricular preexcitation 16%.
- **Variants:** 165 distinct descriptions; cDNA normalized 89.4%; 213 interpretations in the phenopackets (frameshift 34%, nonsense 18%, splice / intronic 15%).
- **Gatekeeper:** 0.61% (110/17,963; exact 95% CI 0.50–0.74) unsupported; 23 of 83 papers; top field `phenotypic_features.status` (25).
- **Repeat (9 papers):** checklist 0.996; variant keys 0.86; same individual count in 7 of 9.
- **ClinVar:** HGVS recall tier 1 35.0% (7/20; 95% CI 18.1–56.7), tiers 1+2 50.0% (10/20; 95% CI 29.9–70.1); overlap 47.9% (35/73; 95% CI 36.9–59.2); classification agreement 78.9% (15/19; 95% CI 56.7–91.5); HPO annotation recall 87.1% (27/31; 95% CI

71.1–94.9). O2 misses: 10 in 5 papers, led by 37288668 (3).

- **Gap candidates:** 35 of 73 tier-1 alleles (27 in one paper, 6 in two, 2 in three or more); 7 only from ClinVar-cited papers; protein check differs for 2; 5 reclassification candidates (1 strong).
- **Input problems flagged by the helpers:** 23571262, 30929317, 42366573: tables transposed or garbled; 25228319: Supplemental Tables 1–2 missing.
- **Highest gatekeeper unsupported share (0.61%)**, with three papers holding 60 of 110: relatives' variant fields from the proband's sentence (37215540), and cohort wording ("usually mild", "typically showed no symptoms") applied to individuals.
- **Sex difference reproduced** (§3.11): males more often show myopathy and hypertrophy; a reproduction check, since the papers overlap with the published series.
- **Variant spectrum:** frameshift 34%, nonsense 18%, splice 15%; LAMP2B-numbered variants (exon 9B) map to intronic positions on the MANE transcript and need the panel's attention (G-danon-002).
- **Input problems:** three papers with transposed or garbled tables and one missing supplement (Supplementary Table S5). Comparator Khedro 2025 is fetched; Boucek 2011 is paywalled.

3.13 Cost and efficiency so far

Table 15. Spend by budget line. Estimates from this session's own cost readings (list price, including helper agents), split into dated stretches and booked to lines (`analysis/cost/spend_segments.csv`, `budget_lines.py`). They are not billing figures: readings sometimes froze for an hour, stages overlapped, and whether this session runs on the funded account is still open (D-15). Other sessions (the pilot-design session) are not included.

Line	Budget line	Requested	Spent (estimate)	Share
1	Extraction across six diseases, incl. re-runs as HGVS and negation handling stabilize	$8,000	$1,132.12	14.2%
2	Gatekeeper verification pass	$4,000	$540.07	13.5%
3	ClinVar/OMIM reconciliation + concordance harness (full re-runs)	$6,000	$64.10	1.1%
4	Claude Code: pipeline, VariantValidator integration, dedup tooling, adjudication interface	$6,000	$15.46	0.3%
5	Ascertainment-bias arm	$4,000	$5.34	0.1%
6	Claude vs open-weight head-to-head	$2,000	$0.00	0.0%
7	Stretch: extend to all 12 candidate diseases	$9,000	$0.00	0.0%
8	Contingency	$6,000	$0.00	0.0%
—	Not yet assigned to a line (setup, documents, decisions, gaps between runs)		$103.97	
	Total, this session's readings	$45,000	**$1,861.06**	4.1%

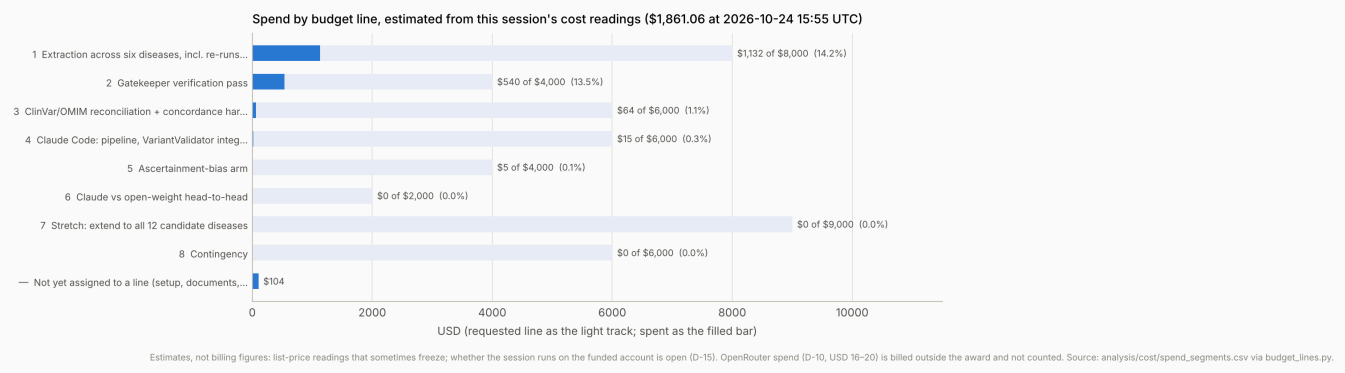


Figure 10. Spend by budget line against the requested amounts.

- **Unit costs measured on this route:** extraction about $2 per paper (helper tokens; $1.08–1.19 median per paper on the two-disease comparison); gatekeeper about 0.0044*perjudgedclaim*; *thesix* – *diseaserunwithits*10769 by session readings ($764 by tokens) against a $1,200 ceiling (D-37); HPO binding helpers cost $72 against a $28 estimate because they searched the ontology file repeatedly.
- **Open-weight compute** (D-10) is billed to OpenRouter, outside the award: $20.35 by key usage over both runs, but the key is shared with other projects, so the host-reported $16.34 is the lower bound and $20.35 the upper.
- **Projection:** the core lines are 6% spent one week into Month 2 (2026-10-21 to 2026-11-20) with extraction, gatekeeping and concordance done for six diseases. What remains on the model side (re-runs if O1 fails, the blinded comparison audit's preparation, bias-arm scripts, write-up) is small; the large remaining costs are human hours, which the award does not fund.

4. Discussion

What the data say so far. The pipeline runs end to end on six diseases at the scale the grant projected (350 papers against "roughly 300"), with provenance that holds on every value and repeats that are stable where the schema fixes the vocabulary. It produces 2,836 valid phenopackets, a ClinVar comparison with intervals on every figure, and 215 candidate gaps that two successive ClinVar releases confirm are absent. Three of the directional expectations stated before registration are consistent with what was seen: classification disagreement runs almost entirely in the "authors say P/LP, ClinVar says VUS" direction; proband counting and deduplication are the weak point (9 of 38 repeat papers differ in individual count; 19 unmerged same-person groups; the pilot's 6 false merges); and the extraction misses concentrate in papers ClinVar cites for variants they do not describe.

What the data do not say. Nothing here establishes the accuracy of the frozen pipeline. The gatekeeper's 0.43% is a filter's reading; the only human-rated figures (6.0%, 6.3%) come from the pilot, before the two rules those audits produced were frozen. The pre-registered test is at most 4 unsupported of 250 under rule C, and the audit's own operating characteristics mean that even a pass would leave a true rate near 2% unexcluded. The concordance figures are ClinVar-only; OMIM, half of the registered answer key, has not been assessed. Recall against ClinVar's citations is low, but the registered triage has not been done, so how much of that is extraction error, normalization error or citations that do not describe the variant is unknown. The open-weight comparison has a Claude judge and no blinded audit yet.

What the gatekeeper is good for. Its value is in the records, not the metric: it removed 1,060 claims before the phenopackets were built, and its reasons name the failure modes a person should look for first (relatives' variant fields, hedged wording, cohort statements). O1 will give its accuracy against human labels, which is the number to report.

On the open-weight result. The cheap model found the same variants and left more values without a quote. If the blinded audit confirms the gatekeeper's reading, the practical conclusion is that the scaffold's provenance rules bind differently on different models, and that span-level provenance is the cheapest place to see it.

Scope. Six diseases, open access only, one extraction model, one judge model. These are pilot results on a pre-registered design, not general findings.

5. Gaps and a plan to close them

Table 16. Gaps, what each blocks, who acts, and what closing it costs. Costs are rough: model costs use this run's measured unit costs (§3.13); panel hours use 1–3 minutes per audited claim and 10–15 minutes per dossier, which are guesses until the adjudication page logs real minutes.

#	Gap	What it blocks	Who acts and how	Rough cost	Time
G1	**O1 human audit not rated** (primary outcome)	O1 pass/fail; gatekeeper accuracy vs human labels (§7.1b); any accuracy claim	Dr. Lin shares the two rater pages with two panel members (D-11); they rate; a third breaks ties	panel: 250 claims × 2 raters at 1–3 min each ≈ 8–25 h, plus tie-break; model: $0	1–3 weeks after sharing
G2	**OMIM not assessed** (D-12)	the OMIM half of O2 and O3; the gap definition's second clause; OMIM allelic-variant candidates (a deliverable)	Dr. Lin submits the drafted API access request; then one harness run	$0 model; 1–2 days of scripting once access arrives	weeks to months (OMIM's response)
G3	**Gap candidates not adjudicated** (O3 stages 2–7)	approved variants, submission drafts, expert-minutes (O5), the gap-closure deliverable	Dr. Lin shares the adjudication page; the panel reviews 215 dossiers; Dr. Lin signs off; submissions under *curation* (D-13)	panel: 215 × 10–15 min ≈ 36–54 h; model: $0 (dossiers exist); LAD-I's 82 could be triaged first (likely numbering slips) to cut this	Months 4–5 per plan
G4	**O2 miss triage not done**	the five-category breakdown of HGVS-recall misses; whether low recall is extraction error	a panel member or Dr. Lin triages 50 sampled misses on the private page	2–4 h of one person	days
G5	**Blinded Claude vs open-weight audit not drawn**	the registered O5 accuracy measure; the only neutral check on the gatekeeper's reading	set the sample size with D-11; draw by seed; same raters, blind to model	panel: 2 × n claims; for n = 100, ≈ 7–20 h; model: $0	with G1
G6	**Bias arm has no numbers** (O4)	per-disease frequency deviation; the ascertainment-bias deliverable	a person fills four comparator sheets without looking at literature frequencies; then run `frequency_deviation.py`	4 sheets × 1–2 h; model: $0	days once a person is assigned
G7	**Open access only** (D-32): 14–53% of each corpus has full text	completeness denominators; recall of the paywalled literature; the gap count is a lower bound	none within the rules; a decision to add institutional access would be a deviation and a re-run	if reopened: extraction ≈ $2 per paper plus gatekeeper; ~300 more papers ≈ $600–900	not planned

#	Gap	What it blocks	Who acts and how	Rough cost	Time
G8	**Known extraction gaps frozen on purpose**: no hedge rule; relatives' variant fields from the proband's sentence; cohort statements applied to individuals	a share of unsupported claims (Figure 3C); would need a v7 and a deviation	wait for O1; if it fails or shows these classes, propose v7 (D-nn) and re-run	full six-disease re-run ≈ $800–1,000 (extraction + gatekeeper + repeat, from this run's spend)	after O1
G9	**Input conversion problems** in ~10 papers (scrambled, transposed or image-only tables; missing or truncated supplements)	recall in those papers (e.g. SSADH 14635103: 12 of 54 families); several O2 misses	fix the converter / refetch supplements; re-extract those papers under a logged deviation	≈ $2–3 per paper plus gatekeeper ≈ $40–60 for 12 papers	days; needs approval
G10	**Notation and normalization**: AxD cDNA without 'c.', legacy numbering, protein-only strings (65 AxD, 85 HA20)	tier-1 matching (recall, overlap, gap candidates for those variants)	harness-side query corrections as in the pilot's D-26 (never in extraction); report by description type	$0 model; 1–2 days scripting; log as a deviation	days
G11	**19 proposed same-person groups unmerged**; 50-merge hand check (D-35) not done	deduplicated counts (lower bounds on collapse 0.2–3.8%); GPSEA denominators	a person confirms groups and checks the registered 50-merge sample; merging after the freeze is a deviation	2–4 h of one person	days
G12	**Checklist-to-HPO maps unreviewed** (five diseases)	the HPO terms behind 20 × 5 checklist items in the phenopackets	Dr. Lin or the panel reviews `configs/checklist_hpo/`	1–2 h	days
G13	**Memorization probe underpowered** (10 post-cutoff papers)	the before/after comparison in §7.4.3	none now; more post-cutoff papers accrue with time	$0	—
G14	**Expert-minutes not measured** (O5)	efficiency per accepted record	comes with G3	$0	with G3
G15	**Budget not reconciled** (D-15): session readings, not billing	spend per line as promised to the program	Dr. Lin reads Settings > Usage on the funded account and confirms which account the sessions ran on	0.5 h	days
G16	**Pre-registration tags not on GitHub; OSF registration not made**	the public, timestamped record of the freeze	Dr. Lin pushes `prereg-v1.0` and `pipeline-v1.0` (the session cannot) and registers on OSF (`reports/drafts/osf_registration.md`)	0.5 h	days
G17	**Drafted messages unsent** (Monarch, ClinVar maintainers, Anthropic credits question, panel note)	Month-1 outreach commitments; the credits/overage question (D-17)	Dr. Lin sends from `reports/drafts/`	0.5 h	days

Recommended order.

Step	Do	Why first
1	Settle D-11 and share the O1 pages; draw the O5 audit sample at the same time (G1, G5)	unblocks the primary outcome and the only neutral check on the gatekeeper
2	Submit the OMIM request; push the tags; register on OSF; send the drafts (G2, G16, G17)	no cost; long lead times start now
3	Triage the 50 O2 misses and LAD-I's 82 gap candidates first (G4, part of G3)	tells us whether low recall and the biggest gap list are real before the panel spends hours on them
4	Fill the four comparator sheets (G6); review the checklist maps (G12); confirm the merge groups (G11)	small human tasks that each unblock a registered figure
5	Only after O1 reads: decide on v7 and a re-run (G8), the input repairs (G9) and the notation repairs (G10)	changes to the frozen pipeline are deviations and should answer a measured failure, not an expected one
6	Panel adjudication of the remaining candidates (G3), then drafts and sign-off	Months 4–5 per the plan; the dossiers and the page are ready

Items 1–4 cost no model spend and about 60–110 panel hours in total, most of it adjudication. Nothing in items 1–4 touches the frozen pipeline. Item 5 is where deviations would start, and it should wait for O1's number.

6. Limitations of this report

- **No human-rated accuracy for the frozen pipeline.** Every support figure in §3.2 is a model's reading of a model's output.
- **HA20 is in every table.** It is flagged and the five-disease subtotals are given, but readers comparing diseases should remember that HA20's rules were tuned on HA20's papers.
- **ClinVar only.** OMIM, half of the registered answer key, is absent from concordance and from the gap definition.
- **Open access only.** The corpora hold 14–53% of the eligible literature; completeness and recall are measured on what was accessible.
- **Cost figures are session estimates** with frozen readings and overlapping stages, and the open-weight key is shared.
- **Helper-reported problems** (input conversion, self-contradicting papers, likely non-gaps) are model output, recorded as such and not verified by a person.
- **The author of this draft is the pipeline's author.** Claude Code built the pipeline, ran it, and generated this report from its outputs. The numbers are reproducible from the committed files; the framing is not independent of the system it describes,

which is one more reason the human audit decides.

7. Data, code and reproducibility

- **Repository:** `cjimmylin/gp-rarelink`, commit `a5e07fc`. Pre-registration: `preregistration/analysis_plan.md` v1.0 (frozen 2026-10-20, D-36). Decisions and deviations: `DECISIONS.md`. Spend: `BUDGET.md`.
- **Records:** `data/extracted/<disease>/pv1-<disease>/` holds per-paper tables, claims with spans, gatekeeper verdicts, reconciled individuals, normalized variants, HPO bindings and the phenopackets. Full-text papers are not committed.
- **Harness outputs:** `analysis/six_disease/`, `adjudication/`, `analysis/openweight/`, `analysis/bias/`, `analysis/exploratory/`.
- **This report:** `analysis/interim_paper/build.py` (text, tables), `figures.py`, `data.py` (inputs; Supplementary Table S1 lists them); output in `reports/interim_paper/`. Re-run with `uv run python -m analysis.interim_paper.build`.
- **Reference snapshots** live outside the repository and are re-created by `pipeline/harness/snapshot_reference.py --per-gene`, `benchmark_chen2020.py` and `catalogue_grossi2024.py`; their manifests and counts are under `analysis/poc/reference_snapshot/`.
- **Rating and adjudication pages** are private claude.ai artifact pages built by scripts under `adjudication/`; their stores are collected by `collect_o1.py` and `apply_ratings.py`. Nothing has been published (standing instruction, 2026-10-21).

Supplementary material

Companion to the interim report of 2026-10-28, commit `a5e07fc`. Generated by the same script.

Supplementary Table S1. Input files

Key	Path
run_summary	`analysis/six_disease/run_summary_2026-10-20.json`
repeat	`analysis/six_disease/repeat_agreement_2026-10-20.json`
stage4	`analysis/six_disease/stage4_summary_2026-10-20.json`
coverage	`analysis/corpus/coverage_2026-10-20.json`
o2	`analysis/six_disease/o2/o2_<d>.json`
o2_misses	`analysis/six_disease/o2/o2_recall_misses.csv`
o2_variants	`analysis/six_disease/o2/o2_variants.csv`
completeness	`analysis/six_disease/completeness_pv1-<d>.json`
gaps	`adjudication/o3_gaps/summary.json`
gap_candidates	`adjudication/o3_gaps/candidates.csv`
triage	`adjudication/o3_gaps/triage_summary.json`
recheck	`adjudication/o3_gaps/recheck_2026-10-21.json`
dossiers	`adjudication/o3_gaps/check_summary.json`
merges	`adjudication/o3_gaps/proposed_merges_summary.json`
merges_csv	`adjudication/o3_gaps/proposed_merges.csv`
reclass	`adjudication/o3_gaps/reclassification_candidates.csv`
gpsea	`analysis/exploratory/gpsea/results/summary.json`
tangier_share	`analysis/exploratory/tangier_clinvar_share.json`
mem_claude	`analysis/bias/memorization/scores_summary.json`
mem_ow	`analysis/bias/memorization/openweight_scores_summary.json`
mem_sample	`analysis/bias/memorization/sample.csv`
bias	`analysis/bias/frequency_deviation_2026-10-21.json`
comparators	`analysis/bias/comparators_2026-10-21.json`
ow_summary	`analysis/openweight/summary_<d>.json`
ow_gk	`analysis/openweight/gatekeeper_compare.json`
ow_o2	`analysis/openweight/o2/o2_<d>.json`
ow_manifest	`runs/ow-<d>/manifest.json`
pv1_manifest	`runs/pv1-<d>/manifest.json`
gk_summary	`data/extracted/<d>/pv1-<d>/gatekeeper_v4_summary.json`
reconcile	`data/extracted/<d>/pv1-<d>/reconcile_summary.json`
build	`data/extracted/<d>/pv1-<d>/phenopacket_build.json`
phenopackets	`data/extracted/<d>/pv1-<d>/phenopackets/*.json`
clinvar_counts	`analysis/poc/reference_snapshot/2026-10-19/counts[_<GENE>].json`
clinvar_counts_recheck	`analysis/poc/reference_snapshot/2026-10-21/counts_<GENE>.json`
poc_audit	`analysis/poc/audit_metrics.json, audit_metrics_v5.json, audit_metrics_v5_excluding_uncertain.json`
poc_dedup	`analysis/poc/dedup_hand_check_summary_poc-ha20-run-2.json`
poc_completeness	`analysis/poc/completeness_poc-ha20-run-2.json`
o1_sample	`adjudication/o1_audit/sample_key.json`
o2_triage_sample	`adjudication/o2_triage/sample_key.json`
checklists	`configs/checklists/<d>.json and schema_ha20_v5.json`
stage4_cfg	`configs/stage4/<d>.json`
budget_segments	`analysis/cost/spend_segments.csv`
budget_lines	`BUDGET.md (Lines table)`
catalogue_check	`analysis/catalogue/catalogue_check_2026-10-20.json`
smoke_manifest	`runs/poc-ha20-v6-smoke/manifest.json`

Supplementary Table S2. ClinVar counts per gene at the two snapshots

Gene	Release (read 2026-10-19)	Single-gene records	P/LP	VUS	Conflicting	Records naming the disease	Release (read 2026-10-21)	Records	P/LP
TNFAIP3	12 Oct 2026	764	99	375	27	76	19 Oct 2026	764	99
ALDH5A1	12 Oct 2026	914	152	333	53	784	19 Oct 2026	914	152
ABCA1	12 Oct 2026	2,006	85	705	141	288	19 Oct 2026	2,006	85
ITGB2	12 Oct 2026	949	87	345	49	830	19 Oct 2026	949	87
GFAP	12 Oct 2026	627	76	262	36	179	19 Oct 2026	629	76
LAMP2	12 Oct 2026	986	154	292	66	640	19 Oct 2026	986	154

Supplementary Table S3. The 20-item checklists per disease (D-38)

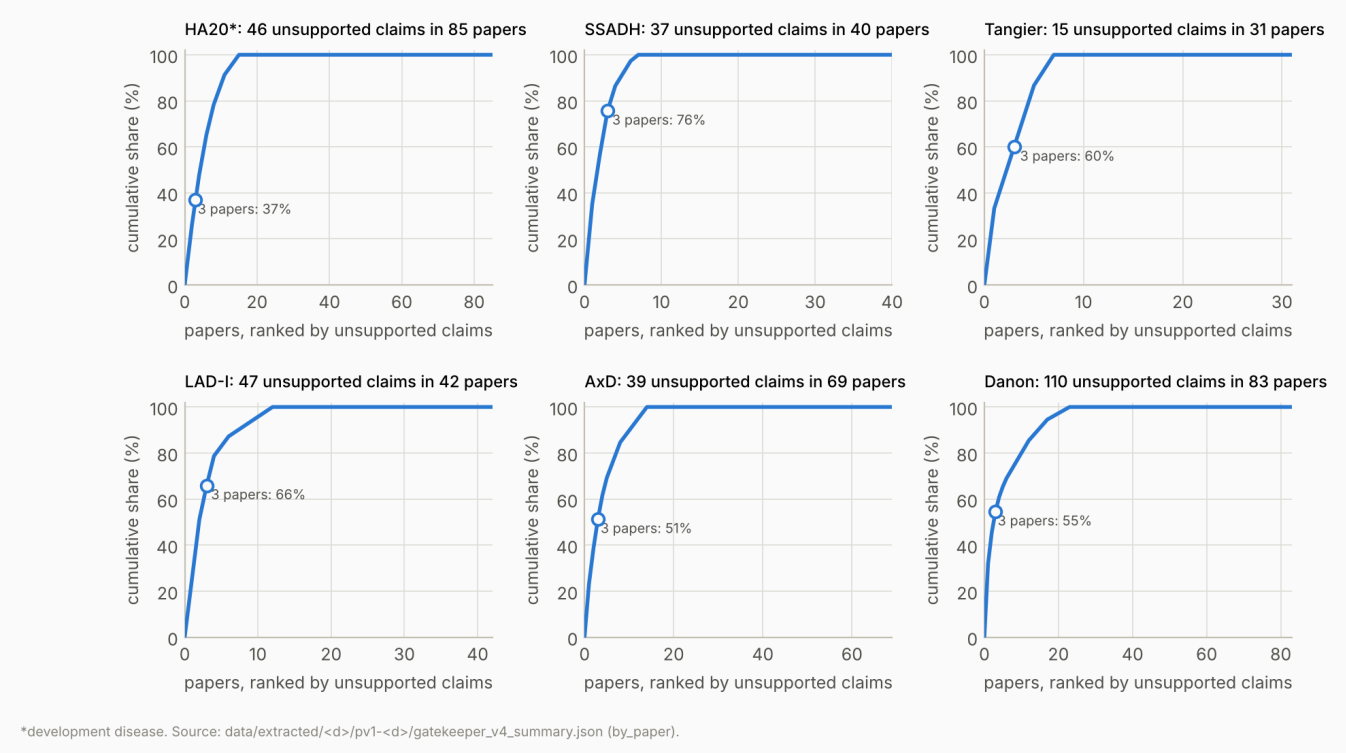
Sources: HA20 pilot schema (schema_ha20_v5.json); SSADH 38452608; Tangier 33994407; LAD-I 42011429; 23207660; 33391282; AxD 39420046; 32779598; 40520824; Danon 37821174.

#	HA20*	SSADH	Tangier	LAD-I	AxD	Danon
1	oral ulcers	intellectual disability	low HDL cholesterol	delayed umbilical cord separation	seizures	left ventricular hypertrophy
2	genital ulcers	speech or language impairment	low apolipoprotein A-I	omphalitis	macrocephaly or megalencephaly	dilated or hypokinetic left ventricle (reduced ejection fraction)
3	gastrointestinal ulcers or inflammation	hypotonia	orange-colored or enlarged tonsils	leukocytosis or neutrophilia	developmental delay	heart failure
4	recurrent fever	gross or fine motor deficits	recurrent tonsillitis or tonsillectomy	absence of pus formation	hydrocephalus or raised intracranial pressure	right ventricular hypertrophy
5	arthritis or arthralgia	seizures or epilepsy	splenomegaly	skin or soft tissue infections	dysphagia	left ventricular outflow tract obstruction
6	uveitis or other ocular inflammation	EEG abnormality (background slowing or epileptiform activity)	hepatomegaly	necrotic skin ulcers or pyoderma gangrenosum-like lesions	dysarthria or dysphonia	late gadolinium enhancement on cardiac MRI
7	skin lesions (rash, folliculitis, erythema nodosum)	ataxia	thrombocytopenia	impaired wound healing	limb weakness	ventricular pre-excitation (short PR, delta wave or Wolff-Parkinson-White)
8	autoimmune thyroiditis	dystonia	reticulocytosis	periodontitis or gingivitis	spasticity or pyramidal signs	atrioventricular block
9	type 1 diabetes	exertional dyskinesia	lymphadenopathy	tooth loss	ataxia or cerebellar signs	supraventricular tachycardia (atrial fibrillation or flutter)
10	autoimmune hepatitis	hemiballismus	corneal opacity	lower respiratory tract infection	palatal tremor	ventricular tachycardia or fibrillation
11	cytopenia (autoimmune)	autism spectrum disorder or autistic behaviors	skin lipid deposits	otitis media	recurrent vomiting	proximal muscle weakness (skeletal myopathy)
12	lymphoproliferation or lymphadenopathy	attention problems or ADHD	intestinal mucosal lipid deposits	sepsis	autonomic dysfunction (orthostatic hypotension, bladder or bowel dysfunction, sleep apnea)	elevated creatine kinase
13	central nervous system involvement	anxiety	peripheral neuropathy	meningitis	cognitive decline	intellectual disability, cognitive impairment or learning difficulties
14	recurrent infections	obsessive-compulsive traits	reduced tendon reflexes	gastrointestinal infections	eye movement abnormality or diplopia	developmental delay (including delayed motor milestones)
15	elevated inflammatory markers (CRP/ESR)	aggression or disruptive behavior	reduced deep sensation	colitis or gastrointestinal inflammatory disease	scoliosis	attention deficit disorder or ADHD
16	autoantibodies (ANA/other)	sleep disturbances	coronary artery disease	diarrhea	cerebral white matter abnormality	mood or anxiety disorder
17	growth failure	depression or mood fluctuation	non-coronary atherosclerotic arterial disease (e.g. brachiocephalic or iliac stenosis)	fungal infections	periventricular rim	pigmentary retinopathy or retinal dystrophy
18	pericarditis or other serositis	psychotic symptoms (hallucinations or delusions)	glucose intolerance or impaired insulin secretion	bcg vaccine complications	basal ganglia or thalamic abnormality	myopia

#	HA20*	SSADH	Tangier	LAD-I	AxD	Danon
19	interstitial lung disease	T2 hyperintensity of globus pallidus, subthalamic or dentate nuclei on MRI	low LDL cholesterol	failure to thrive	contrast enhancement on MRI	elevated liver enzymes (transaminases)
20	vasculitis	cerebral or cerebellar atrophy	hypertriglyceridemia	autoimmune complications	medulla or cervical spinal cord atrophy or signal abnormality	hepatomegaly

Supplementary Table S4. Gatekeeper v4: unsupported claims by field, by whom the quote was about, and the three worst papers

Disease	Top fields (count)	Whom the quote was about	Worst three papers (PMID, unsupported claims)
HA20*	`variants.gene_symbol_as_stated` 17; `phenotypic_features.status` 9; `individuals.clinical_diagnosis_basis` 5; `phenotypic_features.feature_text_as_stated` 3; `variants.protein_hgvs_as_stated` 3	other person 14; this person 10; cohort general 9; named group 9; unclear 4	36325013 (6); 37324276 (6); 31767699 (5)
SSADH	`variants.gene_symbol_as_stated` 16; `variants.transcript_as_stated` 3; `individuals.source_reference_in_review` 3; `variants.genomic_coordinates_as_stated` 3; `phenotypic_features.severity_or_detail` 2	other person 27; this person 10	32055132 (13); 32395407 (8); 21664777 (7)
Tangier	`variants.gene_symbol_as_stated` 9; `phenotypic_features.status` 2; `phenotype_checklist.status` 2; `variants.protein_hgvs_as_stated` 2	other person 11; this person 4	27853448 (5); 38538338 (2); 39096933 (2)
LAD-I	`variants.gene_symbol_as_stated` 14; `variants.protein_hgvs_as_stated` 7; `variants.functional_evidence_statement` 6; `phenotypic_features.status` 5; `individuals.treatment_summary` 4	other person 25; named group 13; this person 7; cohort general 2	36185095 (12); 36353617 (12); 38578558 (7)
AxD	`variants.gene_symbol_as_stated` 21; `phenotypic_features.status` 6; `phenotype_checklist.status` 3; `phenotypic_features.onset_original_text` 2; `variants.protein_hgvs_as_stated` 2	cohort general 18; this person 11; other person 10	27648269 (9); 36088400 (6); 21987397 (5)
Danon	`phenotypic_features.status` 25; `variants.gene_symbol_as_stated` 22; `phenotypic_features.severity_or_detail` 17; `variants.variant_type_as_stated` 9; `variants.variant_original_string` 7	cohort general 55; this person 26; other person 21; named group 6; unclear 2	37215540 (35); 30413001 (15); 41560058 (10)



Supplementary Figure S1. Cumulative share of each disease's unsupported claims against papers ranked by their count.

Supplementary Table S5. Input problems and self-contradicting papers reported by the extraction and gatekeeper helpers

These are helper reports collected in `reports/six_disease_run_findings.md` §5; they are not verified by a person and are not counted in any metric.

Disease	Problem (PMID)
SSADH	14635103: Table 2 scrambled in conversion; about 12 of 54 families extracted
SSADH	38499966: table garbled; supplements S1/S2 missing
SSADH	37503297 and 37962671 are the same study (preprint and journal version); their patients are merged (decision 2026-10-20)
Tangier	26713019 and 27011649: tables published as images
LAD-I	33391282: supplement Table S1 cut at 60,000 characters, so P75–P127 have no clinical rows
AxD	34046764: Table 1 caption only
AxD	32126152: Table S1 missing
AxD	39420046 (Grossi 2024): supplement arrived after extraction (D-39)
Danon	23571262, 30929317, 42366573: tables transposed or garbled
Danon	25228319: Supplemental Tables 1–2 missing

Papers that contradict themselves (table vs text, abstract vs body), examples: LAD-I 36185095: c.920C>T vs c.920T>C; Danon 42382315: DCM vs end-stage HCM for P3; SSADH 37503297/37962671: G176R vs G176E; Danon 30413001: text and Table 1 disagree on which family carries c.56T>C; SSADH 9093300: A.F.'s EEG normal in the table, diffusely slow in the text; AxD/LAD-I/Danon: about 30 papers in all, listed in the run notes for the audit.

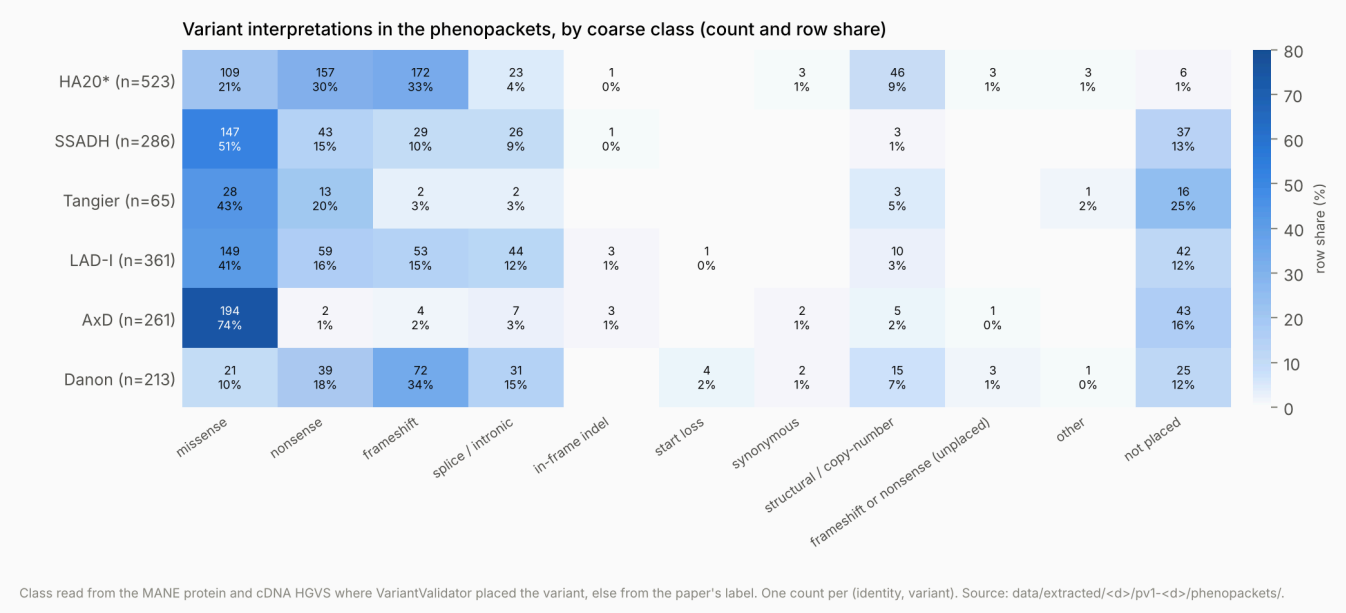
Supplementary Table S6. Run-to-run repeat, per paper

Disease	PMID	Individuals (run / repeat)	Variant keys	Variant strings	Checklist agreement (cells)	Features, label-free
HA20*	29682366	4 / 4	1.00	1.00	0.988 (80)	0.45
HA20*	30402268	4 / 4	1.00	1.00	1.000 (80)	0.75
HA20*	31175876	6 / 7	1.00	0.33	1.000 (120)	0.66
HA20*	32743991	4 / 4	1.00	1.00	1.000 (80)	0.66
HA20*	34011076	32 / 32	0.85	0.37	0.995 (640)	0.87
HA20*	36467491	7 / 7	1.00	1.00	1.000 (140)	0.60
HA20*	40822695	17 / 19	1.00	1.00	0.988 (80)	0.65
HA20*	41226818	1 / 1	1.00	1.00	1.000 (20)	0.95
HA20*	41654260	3 / 3	1.00	1.00	1.000 (60)	1.00
SSADH	32172518	1 / 1	—	—	1.000 (20)	0.18
SSADH	33531951	1 / 1	1.00	1.00	1.000 (20)	0.85
SSADH	38752093	3 / 3	1.00	1.00	1.000 (60)	0.38
SSADH	38791277	3 / 3	1.00	1.00	1.000 (60)	0.33
Tangier	33346003	1 / 1	—	1.00	1.000 (20)	0.42
Tangier	37545679	3 / 3	1.00	1.00	0.950 (20)	0.48
Tangier	38538338	6 / 4	0.75	1.00	0.975 (80)	0.72
Tangier	39096933	4 / 4	1.00	0.00	0.963 (80)	0.59
LAD-I	24995649	4 / 4	1.00	0.33	1.000 (80)	0.53
LAD-I	30312521	1 / 1	1.00	1.00	1.000 (20)	0.40
LAD-I	36185095	7 / 6	1.00	0.50	1.000 (120)	0.81
LAD-I	36353617	5 / 5	1.00	0.67	1.000 (100)	0.62
LAD-I	42625796	8 / 6	0.75	0.50	0.992 (120)	0.71
AxD	20359319	6 / 6	1.00	0.00	0.992 (120)	0.57
AxD	21987397	28 / 28	1.00	1.00	0.995 (420)	0.85
AxD	31611638	31 / 31	1.00	1.00	1.000 (620)	1.00
AxD	35040226	2 / 2	1.00	1.00	1.000 (40)	0.38
AxD	39831714	1 / 1	—	—	0.950 (20)	0.67
AxD	40350261	4 / 4	1.00	1.00	1.000 (80)	0.77
AxD	40452725	2 / 2	—	1.00	1.000 (40)	0.79
Danon	20173215	3 / 3	0.50	0.50	1.000 (60)	0.52
Danon	25228319	7 / 7	1.00	1.00	1.000 (120)	0.60

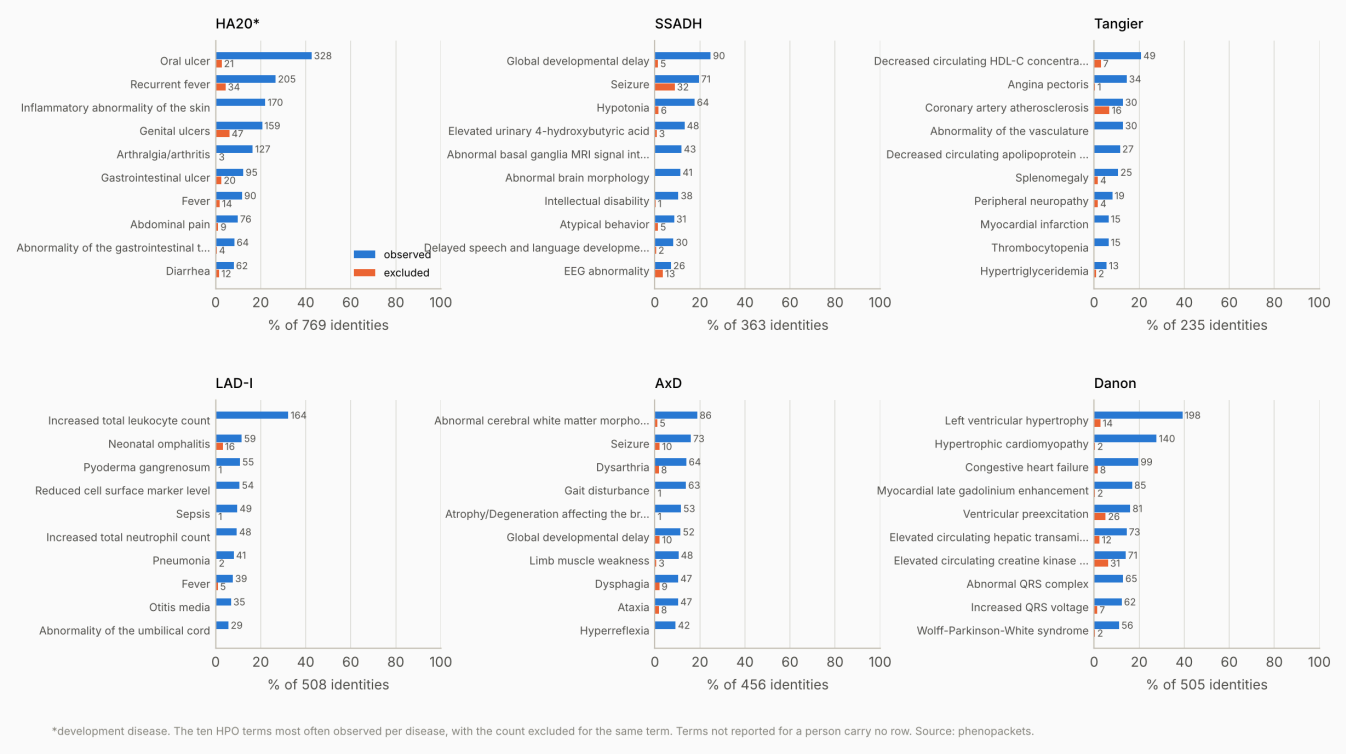
Disease	PMID	Individuals (run / repeat)	Variant keys	Variant strings	Checklist agreement (cells)	Features, label-free
Danon	28289499	1 / 1	—	1.00	1.000 (20)	0.77
Danon	29637036	3 / 3	1.00	1.00	1.000 (60)	0.58
Danon	31709377	1 / 3	1.00	0.00	1.000 (20)	0.70
Danon	34362124	4 / 4	1.00	0.33	1.000 (80)	0.35
Danon	37609033	12 / 11	1.00	1.00	0.996 (220)	0.66
Danon	38256360	4 / 4	1.00	1.00	0.975 (80)	0.62
Danon	38606381	1 / 1	1.00	1.00	1.000 (20)	0.50

Supplementary Table S7. Stage 4 details: normalization status, checklist-derived terms and claims the gatekeeper removed

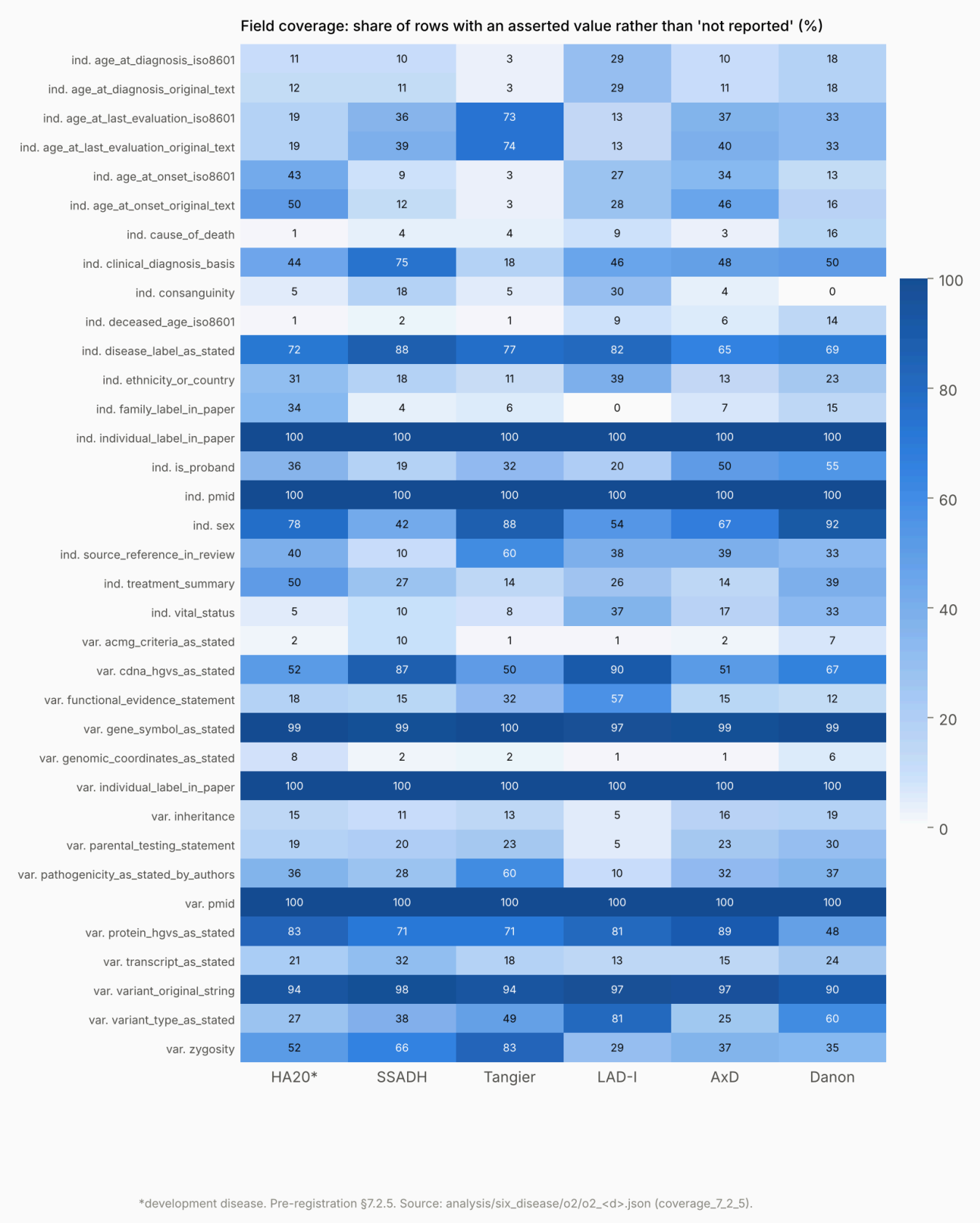
Disease	Normalized	Corrected query	Protein-only	Not normalized	Unresolvable	MANE Select	Feature rows unbound (skipped)	Checklist terms observed	Checklist terms excluded	Claims dropped by gatekeeper, by kind
HA20*	106	41	85	43	5	NM_001270508.2	424	452	175	age 22; checklist_rows 90; disease_label 3; feature_rows 30; onset 34; pathogenicity 2; variant_rows 24; zygosity 2
SSADH	106	16	15	6	19	NM_001080.3	172	109	38	age 16; checklist_rows 17; feature_rows 21; onset 5; pathogenicity 1; sex 1; variant_rows 90
Tangier	18	0	17	11	2	NM_005502.4	98	45	29	age 75; checklist_rows 12; disease_label 1; feature_rows 17; onset 2; variant_rows 10
LAD-I	112	35	7	15	23	NM_000211.5	293	26	36	age 15; checklist_rows 9; disease_label 16; feature_rows 16; onset 15; variant_rows 27; zygosity 2
AxD	80	2	65	12	22	NM_002055.5	288	127	69	age 61; checklist_rows 37; disease_label 1; feature_rows 42; onset 16; pathogenicity 4; variant_rows 60
Danon	62	39	25	27	12	NM_002294.3	328	191	210	age 18; checklist_rows 79; disease_label 3; feature_rows 88; onset 29; pathogenicity 3; variant_rows 41; zygosity 3



Supplementary Figure S2. Variant interpretations in the phenopackets by coarse class.



Supplementary Figure S3. The ten most-observed HPO terms per disease, with the count excluded for the same term.



Supplementary Figure S4. Field coverage: the share of rows with an asserted value rather than 'not reported'.

Supplementary Table S8. O2 details

Classification confusion (ClinVar aggregate | authors' assertion: count).

Disease	Cells
HA20*	B/LB

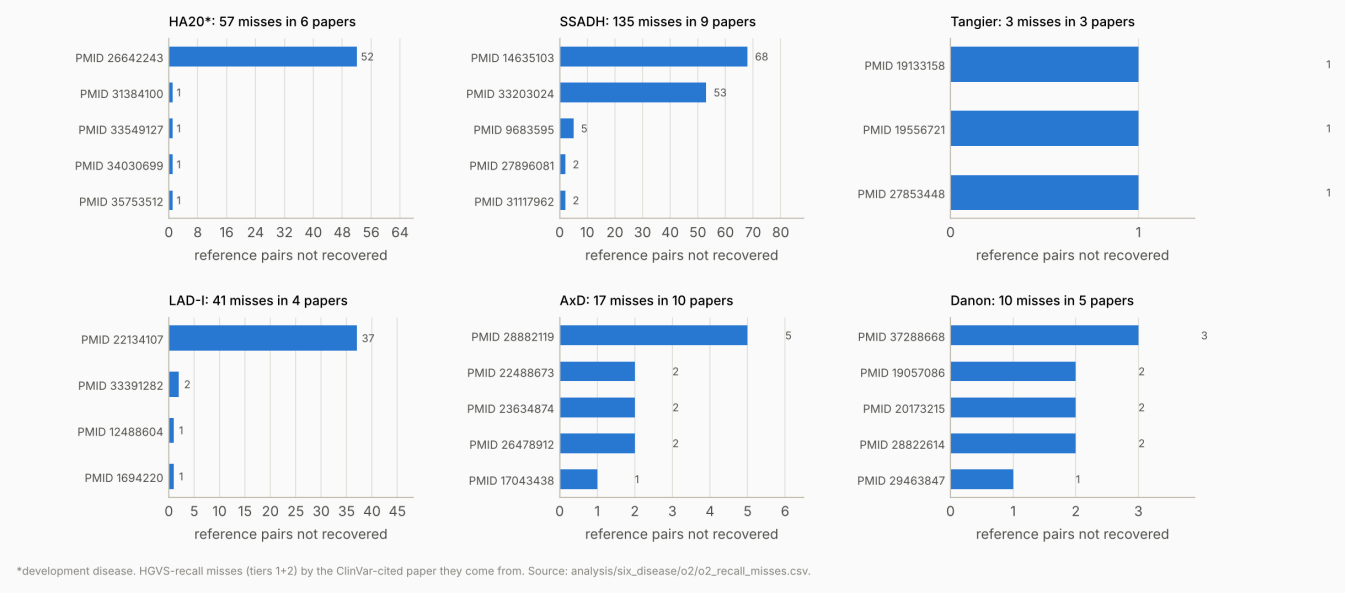
Disease	Cells
SSADH	B/LB
Tangier	B/LB
LAD-I	P/LP
AxD	B/LB
Danon	P/LP

Extracted variants by match tier.

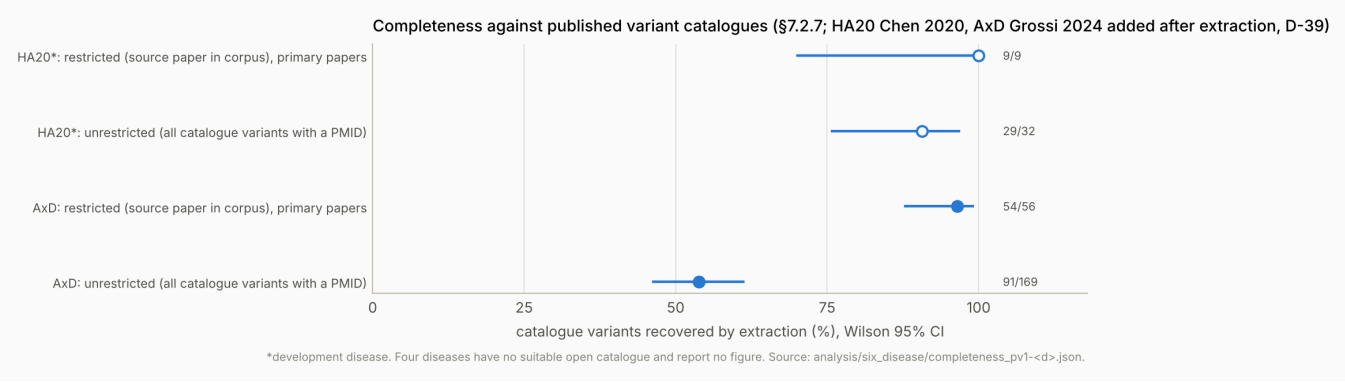
Disease	Tier 1 (allele)	Tier 2 (protein)	Unmatched (placed)	Unmatched (protein-only)
HA20*	39	18	64	69
SSADH	68	16	10	4
Tangier	9	7	7	12
LAD-I	45	7	84	16
AxD	49	62	18	12
Danon	35	13	38	17

HGVS-recall misses by ClinVar-cited paper.

Disease	PMID	Reference pairs not recovered
HA20*	26642243	52
HA20*	31384100	1
HA20*	33549127	1
HA20*	34030699	1
HA20*	35753512	1
HA20*	35958611	1
SSADH	14635103	68
SSADH	33203024	53
SSADH	9683595	5
SSADH	27896081	2
SSADH	31117962	2
SSADH	38791277	2
SSADH	30829465	1
SSADH	32395407	1
SSADH	37962671	1
Tangier	19133158	1
Tangier	19556721	1
Tangier	27853448	1
LAD-I	22134107	37
LAD-I	33391282	2
LAD-I	12488604	1
LAD-I	1694220	1
AxD	28882119	5
AxD	22488673	2
AxD	23634874	2
AxD	26478912	2
AxD	17043438	1
AxD	19672978	1
AxD	27814755	1
AxD	29057636	1
AxD	34012265	1
AxD	37396762	1
Danon	37288668	3
Danon	19057086	2
Danon	20173215	2
Danon	28822614	2
Danon	29463847	1



Supplementary Figure S7. O2 misses by paper.



Supplementary Figure S5. Completeness against the two published catalogues.

Supplementary Table S9. Gap candidates, reclassification candidates and proposed same-person groups

All 215 gap candidates (`adjudication/o3_gaps/candidates.csv`; dossiers in `dossiers/`). Pending adjudication; OMIM not checked.

Candidate	cDNA (MANE)	Predicted protein	Protein check	Papers	Identities	Authors' class	Only ClinVar-cited papers
G-ha20-001	NM_001270508.2:c.252del	NP_001257437.1:p.(Trp85GlyfsTer11)	agrees	3	5	P/LP	False
G-ha20-002	NM_001270508.2:c.271A>T	NP_001257437.1:p.(Lys91Ter)	agrees	2	7	P/LP	False
G-ha20-003	NM_001270508.2:c.296-2A>G	NP_001257437.1:p.?	no stated protein	1	1	not stated	False
G-ha20-004	NM_001270508.2:c.460G>T	NP_001257437.1:p.(Glu154Ter)	agrees	1	3	not stated	False
G-ha20-005	NM_001270508.2:c.487-1G>A	NP_001257437.1:p.?	differs (stated N163fs, predicted ?)	1	1	P/LP	False
G-ha20-006	NM_001270508.2:c.492G>A	NP_001257437.1:p.(Trp164Ter)	agrees	2	3	P/LP	False
G-ha20-007	NM_001270508.2:c.520del	NP_001257437.1:p.(Met174TrpfsTer42)	agrees	1	1	P/LP	False
G-ha20-008	NM_001270508.2:c.559C>T	NP_001257437.1:p.(Gln187Ter)	agrees	3	4	P/LP	False
G-ha20-009	NM_001270508.2:c.574G>A	NP_001257437.1:p.(Glu192Lys)	agrees	3	5	P/LP	False
G-ha20-010	NM_001270508.2:c.608T>G	NP_001257437.1:p.(Leu203Arg)	agrees	1	2	P/LP	False

Candidate	cDNA (MANE)	Predicted protein	Protein check	Papers	Identities	Authors' class	Only ClinVar-cited papers
G-ha20-011	NM_001270508.2:c.610del	NP_001257437.1:p.(Arg204GlufsTer12)	agrees	1	1	P/LP	False
G-ha20-012	NM_001270508.2:c.634+2T>C	NP_001257437.1:p.?	differs (stated D212fs, predicted ?)	2	1	P/LP	False
G-ha20-013	NM_001270508.2:c.635-1G>A	NP_001257437.1:p.?	no stated protein	1	3	not stated	False
G-ha20-014	NM_001270508.2:c.677del	NP_001257437.1:p.(Pro226LeufsTer2)	agrees	1	1	P/LP	False
G-ha20-015	NM_001270508.2:c.707T>C	NP_001257437.1:p.(Leu236Pro)	agrees	2	4	mixed	False
G-ha20-016	NM_001270508.2:c.728G>A	NP_001257437.1:p.(Cys243Tyr)	agrees	3	9	P/LP	False
G-ha20-017	NM_001270508.2:c.759del	NP_001257437.1:p.(Asp253GlufsTer9)	agrees	1	1	P/LP	False
G-ha20-018	NM_001270508.2:c.805+1G>A	NP_001257437.1:p.?	differs (stated N212fs, predicted ?)	1	4	P/LP	False
G-ha20-019	NM_001270508.2:c.806-1G>T	NP_001257437.1:p.?	no stated protein	1	1	not stated	False
G-ha20-020	NM_001270508.2:c.821del	NP_001257437.1:p.(Pro274HisfsTer13)	agrees	2	4	P/LP	False
G-ha20-021	NM_001270508.2:c.866del	NP_001257437.1:p.(His289ProfsTer3)	agrees	2	2	P/LP	False
G-ha20-022	NM_001270508.2:c.925_927dup	NP_001257437.1:p.(Val309dup)	agrees	1	1	P/LP	True
G-ha20-023	NM_001270508.2:c.986+1G>C	NP_001257437.1:p.?	differs (stated I310fs, predicted ?)	1	1	P/LP	False
G-ha20-024	NM_001270508.2:c.986+1G>T	NP_001257437.1:p.?	differs (stated K329N*1, predicted ?)	1	1	P/LP	False
G-ha20-025	NM_001270508.2:c.994G>T	NP_001257437.1:p.(Glu332Ter)	agrees	2	4	not stated	False
G-ha20-026	NM_001270508.2:c.997dup	NP_001257437.1:p.(Ala333GlyfsTer2)	agrees	1	1	P/LP	False
G-ha20-027	NM_001270508.2:c.1012G>T	NP_001257437.1:p.(Glu338Ter)	agrees	1	1	not stated	False
G-ha20-028	NM_001270508.2:c.1067G>A	NP_001257437.1:p.(Trp356Ter)	agrees	1	1	not stated	False
G-ha20-029	NM_001270508.2:c.1128del	NP_001257437.1:p.(Val377CysfsTer9)	differs (stated S376fs, predicted V377fs)	1	1	P/LP	False
G-ha20-030	NM_001270508.2:c.1217A>T	NP_001257437.1:p.(Glu406Val)	agrees	1	1	not stated	False
G-ha20-031	NM_001270508.2:c.1223C>G	NP_001257437.1:p.(Ser408Ter)	agrees	1	1	P/LP	False
G-ha20-032	NM_001270508.2:c.1247_1251del	NP_001257437.1:p.(Asn416ThrfsTer11)	agrees	1	1	P/LP	False
G-ha20-033	NM_001270508.2:c.1249_1252del	NP_001257437.1:p.(Lys417SerfsTer4)	agrees	3	3	P/LP	False
G-ha20-034	NM_001270508.2:c.1300_1301delinsTA	NP_001257437.1:p.(Ala434Ter)	agrees	3	2	P/LP	False
G-ha20-035	NM_001270508.2:c.1309del	NP_001257437.1:p.(Ala437ProfsTer40)	agrees	1	1	not stated	False
G-ha20-036	NM_001270508.2:c.1318_1319del	NP_001257437.1:p.(Gly440ArgfsTer4)	agrees	1	1	P/LP	True
G-ha20-037	NM_001270508.2:c.1420A>G	NP_001257437.1:p.(Thr474Ala)	agrees	1	1	VUS	False
G-ha20-038	NM_001270508.2:c.1434C>A	NP_001257437.1:p.(Cys478Ter)	agrees	1	2	P/LP	False
G-ha20-039	NM_001270508.2:c.1466_1467del	NP_001257437.1:p.(Val489AlafsTer7)	agrees	1	1	not stated	False
G-ha20-040	NM_001270508.2:c.1467_1468delinsAT	NP_001257437.1:p.(Gln490Ter)	agrees	2	1	P/LP	False
G-ha20-041	NM_001270508.2:c.1518del	NP_001257437.1:p.(Ser507AlafsTer190)	agrees	1	1	P/LP	False
G-ha20-042	NM_001270508.2:c.1631dup	NP_001257437.1:p.(Ala545SerfsTer127)	agrees	1	1	not stated	False
G-ha20-043	NM_001270508.2:c.1639G>A	NP_001257437.1:p.(Ala547Thr)	agrees	2	7	mixed	True

Candidate	cDNA (MANE)	Predicted protein	Protein check	Papers	Identities	Authors' class	Only ClinVar-cited papers
G-ha20-044	NM_001270508.2:c.1690A>T	NP_001257437.1:p.(Lys564Ter)	agrees	1	1	not stated	False
G-ha20-045	NM_001270508.2:c.1733C>T	NP_001257437.1:p.(Ser578Phe)	agrees	1	2	VUS	False
G-ha20-046	NM_001270508.2:c.1747G>T	NP_001257437.1:p.(Gly583Ter)	agrees	2	2	P/LP	False
G-ha20-047	NM_001270508.2:c.1763_1773del	NP_001257437.1:p.(Ala588ValfsTer80)	agrees	1	2	P/LP	False
G-ha20-048	NM_001270508.2:c.1804A>T	NP_001257437.1:p.(Thr602Ser)	agrees	1	3	P/LP	True
G-ha20-049	NM_001270508.2:c.1807G>A	NP_001257437.1:p.(Gly603Arg)	agrees	1	1	VUS	True
G-ha20-050	NM_001270508.2:c.1831_1835dup	NP_001257437.1:p.(Cys612TrpfsTer87)	agrees	1	1	P/LP	False
G-ha20-051	NM_001270508.2:c.1903_1906del	NP_001257437.1:p.(Lys635IlefsTer61)	agrees	1	1	P/LP	False
G-ha20-052	NM_001270508.2:c.1906+1G>C	NP_001257437.1:p.?	no stated protein	2	1	P/LP	False
G-ha20-053	NM_001270508.2:c.1906+2T>G	NP_001257437.1:p.?	no stated protein	2	2	P/LP	False
G-ha20-054	NM_001270508.2:c.2088+1G>A	NP_001257437.1:p.?	differs (stated H636fs, predicted ?)	1	1	P/LP	False
G-ha20-055	NM_001270508.2:c.2088+5G>C	NP_001257437.1:p.?	differs (stated H636fs, predicted ?)	2	2	P/LP	False
G-ha20-056	NM_001270508.2:c.2093C>A	NP_001257437.1:p.(Ser698Ter)	agrees	1	1	not stated	False
G-ha20-057	NM_001270508.2:c.2126A>G	NP_001257437.1:p.(Gln709Arg)	agrees	2	2	B/LB	True
G-ha20-058	NM_001270508.2:c.2152del	NP_001257437.1:p.(Arg718GlyfsTer98)	no stated protein	1	1	not stated	False
G-ha20-059	NM_001270508.2:c.2199C>A	NP_001257437.1:p.(Cys733Ter)	agrees	1	1	VUS	False
G-ha20-060	NM_001270508.2:c.2209del	NP_001257437.1:p.(Gln737SerfsTer79)	agrees	4	19	P/LP	False
G-ha20-061	NM_001270508.2:c.2251G>T	NP_001257437.1:p.(Glu751Ter)	agrees	1	1	P/LP	False
G-ha20-062	NM_001270508.2:c.2274del	NP_001257437.1:p.(Lys759SerfsTer57)	agrees	1	1	P/LP	False
G-ha20-063	NM_001270508.2:c.2300G>A	NP_001257437.1:p.(Cys767Tyr)	agrees	1	1	P/LP	False
G-ha20-064	NM_001270508.2:c.2317_2318del	NP_001257437.1:p.(Ala773GlnfsTer57)	agrees	1	2	P/LP	False
G-ssadh-001	NM_001080.3:c.127del	NP_001071.1:p.(Gln43SerfsTer48)	agrees	2	1	not stated	False
G-ssadh-002	NM_001080.3:c.400_401del	NP_001071.1:p.(Asn134Ter)	differs (stated Q134*, predicted N134*)	1	1	P/LP	False
G-ssadh-003	NM_001080.3:c.610-612T>G	NP_001071.1:p.?	no stated protein	1	1	P/LP	True
G-ssadh-004	NM_001080.3:c.644_647del	NP_001071.1:p.(Val215GlyfsTer12)	agrees	2	1	not stated	False
G-ssadh-005	NM_001080.3:c.728T>C	NP_001071.1:p.(Leu243Pro)	agrees	1	1	P/LP	True
G-ssadh-006	NM_001080.3:c.755G>A	NP_001071.1:p.(Gly252Asp)	agrees	2	1	not stated	False
G-ssadh-007	NM_001080.3:c.865G>A	NP_001071.1:p.(Gly289Arg)	agrees	1	1	P/LP	False
G-ssadh-008	NM_001080.3:c.870+3del	NP_001071.1:p.?	no stated protein	2	1	not stated	False
G-ssadh-009	NM_001080.3:c.983C>A	NP_001071.1:p.(Ala328Asp)	agrees	1	1	P/LP	False
G-ssadh-010	NM_001080.3:c.1105C>G	NP_001071.1:p.(Arg369Gly)	agrees	1	1	P/LP	False
G-tangier-001	NM_005502.4:c.5094C>A	NP_005493.2:p.(Tyr1698Ter)	agrees	1	1	not stated	True

Candidate	cDNA (MANE)	Predicted protein	Protein check	Papers	Identities	Authors' class	Only ClinVar-cited papers
G-tangier-002	NM_005502.4:c.5055del	NP_005493.2:p.(Lys1685AsnfsTer9)	agrees	1	1	P/LP	False
G-tangier-003	NM_005502.4:c.4799A>G	NP_005493.2:p.(His1600Arg)	agrees	1	3	P/LP	False
G-tangier-004	NM_005502.4:c.4297G>T	NP_005493.2:p.(Glu1433Ter)	agrees	1	1	not stated	False
G-tangier-005	NM_005502.4:c.3460A>T	NP_005493.2:p.(Lys1154Ter)	agrees	1	1	P/LP	False
G-tangier-006	NM_005502.4:c.2418G>A	NP_005493.2:p.(Trp806Ter)	no stated protein	1	2	P/LP	False
G-tangier-007	NM_005502.4:c.1834G>T	NP_005493.2:p.(Glu612Ter)	differs (stated - E612*, predicted E612*)	1	1	not stated	False
G-lad1-001	NM_000211.5:c.2147G>T	NP_000202.3:p.(Gly716Val)	agrees	1	1	not stated	True
G-lad1-002	NM_000211.5:c.1413-149_2080+839delinsAAAA	NP_000202.3:p.?	differs (stated C472fs, predicted ?)	1	1	not stated	True
G-lad1-003	NM_000211.5:c.2080+1del	NP_000202.3:p.?	differs (stated S627fs, predicted ?)	1	1	not stated	True
G-lad1-004	NM_000211.5:c.2055C>A	NP_000202.3:p.(Tyr685Ter)	differs (stated Try685*, predicted Y685*)	1	1	not stated	True
G-lad1-005	NM_000211.5:c.2003_2004del	NP_000202.3:p.(Glu668GlyfsTer23)	agrees	1	1	not stated	False
G-lad1-006	NM_000211.5:c.1920del	NP_000202.3:p.(Lys641ArgfsTer17)	agrees	1	1	not stated	True
G-lad1-007	NM_000211.5:c.1878-1G>A	NP_000202.3:p.?	no stated protein	1	1	not stated	True
G-lad1-008	NM_000211.5:c.1878-2A>C	NP_000202.3:p.?	no stated protein	1	2	not stated	True
G-lad1-009	NM_000211.5:c.1866T>A	NP_000202.3:p.(Cys622Ter)	agrees	1	1	not stated	True
G-lad1-010	NM_000211.5:c.1848del	NP_000202.3:p.(Gly617AlafsTer16)	agrees	1	1	not stated	False
G-lad1-011	NM_000211.5:c.1840G>T	NP_000202.3:p.(Glu614Ter)	agrees	1	2	not stated	True
G-lad1-012	NM_000211.5:c.1834T>C	NP_000202.3:p.(Cys612Arg)	agrees	1	1	not stated	True
G-lad1-013	NM_000211.5:c.1828C>A	NP_000202.3:p.(Pro610Thr)	agrees	1	1	not stated	True
G-lad1-014	NM_000211.5:c.1798del	NP_000202.3:p.(Val600TyrfsTer33)	agrees	1	1	not stated	True
G-lad1-015	NM_000211.5:c.1745G>A	NP_000202.3:p.(Cys582Tyr)	agrees	1	1	not stated	True
G-lad1-016	NM_000211.5:c.1670G>C	NP_000202.3:p.(Cys557Ser)	agrees	1	1	not stated	True
G-lad1-017	NM_000211.5:c.1670G>A	NP_000202.3:p.(Cys557Tyr)	agrees	1	1	not stated	False
G-lad1-018	NM_000211.5:c.1663G>A	NP_000202.3:p.(Gly555Arg)	agrees	1	1	not stated	False
G-lad1-019	NM_000211.5:c.1658-2A>G	NP_000202.3:p.?	differs (stated G553fs, predicted ?)	2	2	not stated	True
G-lad1-020	NM_000211.5:c.1628_1657+6del	NP_000202.3:p.?	differs (stated C541_G5, predicted ?)	1	1	not stated	True
G-lad1-021	NM_000211.5:c.1632C>G	NP_000202.3:p.(Tyr544Ter)	agrees	2	4	not stated	True
G-lad1-022	NM_000211.5:c.1621T>C	NP_000202.3:p.(Cys541Arg)	agrees	1	1	not stated	True
G-lad1-023	NM_000211.5:c.1590C>G	NP_000202.3:p.(Tyr530Ter)	agrees	2	2	not stated	True
G-lad1-024	NM_000211.5:c.1516del	NP_000202.3:p.(Cys506AlafsTer23)	agrees	1	1	not stated	True
G-lad1-025	NM_000211.5:c.1498del	NP_000202.3:p.(Asp500ThrfsTer29)	agrees	1	6	not stated	True
G-lad1-026	NM_000211.5:c.1472_1475del	NP_000202.3:p.(Gln491ArgfsTer37)	agrees	1	1	not stated	True
G-lad1-027	NM_000211.5:c.1421del	NP_000202.3:p.(Thr474MetfsTer55)	agrees	1	1	not stated	True
G-lad1-028	NM_000211.5:c.1388_1390delinsCA	NP_000202.3:p.(Gly463AlafsTer66)	agrees	1	1	not stated	True
G-lad1-029	NM_000211.5:c.1283T>G	NP_000202.3:p.(Ile428Ser)	agrees	1	1	not stated	True
G-lad1-030	NM_000211.5:c.1256_1257del	NP_000202.3:p.(Glu419ValfsTer27)	agrees	1	1	not stated	True
G-lad1-031	NM_000211.5:c.1224+4A>G	NP_000202.3:p.?	no stated protein	1	5	not stated	True
G-lad1-032	NM_000211.5:c.1083+5G>C	NP_000202.3:p.?	differs (stated K332_N3, predicted ?)	1	1	not stated	True

Candidate	cDNA (MANE)	Predicted protein	Protein check	Papers	Identities	Authors' class	Only ClinVar-cited papers
G-lad1-033	NM_000211.5:c.1083+3G>C	NP_000202.3:p.?	differs (stated K332_N3, predicted ?)	1	5	not stated	True
G-lad1-034	NM_000211.5:c.1021G>C	NP_000202.3:p.(Ala341Pro)	agrees	1	1	not stated	True
G-lad1-035	NM_000211.5:c.995_1004del	NP_000202.3:p.(Lys332ArgfsTer44)	agrees	1	1	not stated	True
G-lad1-036	NM_000211.5:c.962C>A	NP_000202.3:p.(Ala321Glu)	agrees	1	1	not stated	True
G-lad1-037	NM_000211.5:c.953C>A	NP_000202.3:p.(Pro318His)	agrees	1	1	not stated	True
G-lad1-038	NM_000211.5:c.943_946dup	NP_000202.3:p.(Ile316LysfsTer11)	agrees	1	2	not stated	True
G-lad1-039	NM_000211.5:c.920T>C	NP_000202.3:p.(Leu307Pro)	agrees	1	1	P/LP	False
G-lad1-040	NM_000211.5:c.905C>T	NP_000202.3:p.(Pro302Leu)	agrees	1	1	not stated	True
G-lad1-041	NM_000211.5:c.897+1G>C	NP_000202.3:p.?	no stated protein	1	1	not stated	True
G-lad1-042	NM_000211.5:c.844_846del	NP_000202.3:p.(Asn282del)	agrees	1	3	P/LP	False
G-lad1-043	NM_000211.5:c.846C>A	NP_000202.3:p.(Asn282Lys)	agrees	1	1	not stated	True
G-lad1-044	NM_000211.5:c.769C>T	NP_000202.3:p.(Arg257Trp)	agrees	1	1	not stated	True
G-lad1-045	NM_000211.5:c.756G>C	NP_000202.3:p.(Trp252Cys)	agrees	1	2	not stated	True
G-lad1-046	NM_000211.5:c.755G>A	NP_000202.3:p.(Trp252Ter)	agrees	1	1	not stated	True
G-lad1-047	NM_000211.5:c.754T>C	NP_000202.3:p.(Trp252Arg)	agrees	1	1	not stated	True
G-lad1-048	NM_000211.5:c.751G>A	NP_000202.3:p.(Gly251Ser)	agrees	1	5	not stated	True
G-lad1-049	NM_000211.5:c.742-1G>A	NP_000202.3:p.?	no stated protein	1	3	not stated	True
G-lad1-050	NM_000211.5:c.725A>G	NP_000202.3:p.(Gln242Arg)	agrees	1	1	not stated	True
G-lad1-051	NM_000211.5:c.712G>A	NP_000202.3:p.(Asp238Asn)	agrees	1	1	not stated	True
G-lad1-052	NM_000211.5:c.710T>G	NP_000202.3:p.(Leu237Arg)	agrees	1	1	not stated	True
G-lad1-053	NM_000211.5:c.691G>C	NP_000202.3:p.(Asp231His)	agrees	1	2	not stated	True
G-lad1-054	NM_000211.5:c.658G>T	NP_000202.3:p.(Glu220Ter)	agrees	1	5	not stated	True
G-lad1-055	NM_000211.5:c.652C>T	NP_000202.3:p.(Gln218Ter)	no stated protein	1	1	not stated	False
G-lad1-056	NM_000211.5:c.614_615insA	NP_000202.3:p.(His206AlafsTer59)	agrees	1	1	not stated	True
G-lad1-057	NM_000211.5:c.602del	NP_000202.3:p.(Pro201ArgfsTer8)	agrees	1	1	not stated	True
G-lad1-058	NM_000211.5:c.557dup	NP_000202.3:p.(Leu187AlafsTer78)	agrees	1	1	not stated	True
G-lad1-059	NM_000211.5:c.520A>G	NP_000202.3:p.(Lys174Glu)	agrees	1	1	not stated	True
G-lad1-060	NM_000211.5:c.500G>T	NP_000202.3:p.(Gly167Val)	agrees	1	1	not stated	False
G-lad1-061	NM_000211.5:c.500-12T>G	NP_000202.3:p.?	differs (stated G167fs, predicted ?)	1	1	not stated	True
G-lad1-062	NM_000211.5:c.499+1G>T	NP_000202.3:p.?	no stated protein	1	1	not stated	True
G-lad1-063	NM_000211.5:c.449G>A	NP_000202.3:p.(Gly150Asp)	agrees	1	1	not stated	True
G-lad1-064	NM_000211.5:c.329-2A>G	NP_000202.3:p.?	no stated protein	1	1	not stated	True
G-lad1-065	NM_000211.5:c.329-6C>A	NP_000202.3:p.?	differs (stated Q111_F1, predicted ?)	1	1	not stated	True
G-lad1-066	NM_000211.5:c.328+1G>A	NP_000202.3:p.?	differs (stated N50fs, predicted ?)	1	1	not stated	True
G-lad1-067	NM_000211.5:c.307del	NP_000202.3:p.(Val103Ter)	no stated protein	1	1	not stated	False
G-lad1-068	NM_000211.5:c.306dup	NP_000202.3:p.(Val103SerfsTer39)	agrees	1	1	not stated	False
G-lad1-069	NM_000211.5:c.268del	NP_000202.3:p.(Asp90ThrfsTer14)	agrees	1	1	not stated	True
G-lad1-070	NM_000211.5:c.262C>T	NP_000202.3:p.(Gln88Ter)	agrees	1	1	P/LP	False
G-lad1-071	NM_000211.5:c.215del	NP_000202.3:p.(Gly72AlafsTer32)	agrees	1	1	not stated	True
G-lad1-072	NM_000211.5:c.199C>T	NP_000202.3:p.(Gln67Ter)	agrees	1	3	not stated	True
G-lad1-073	NM_000211.5:c.185G>A	NP_000202.3:p.(Cys62Tyr)	agrees	1	2	not stated	False
G-lad1-074	NM_000211.5:c.184T>C	NP_000202.3:p.(Cys62Arg)	agrees	1	1	not stated	True
G-lad1-075	NM_000211.5:c.134G>A	NP_000202.3:p.(Trp45Ter)	agrees	1	1	not stated	False
G-lad1-076	NM_000211.5:c.130A>C	NP_000202.3:p.(Thr44Pro)	agrees	2	4	not stated	False
G-lad1-077	NM_000211.5:c.119_128del	NP_000202.3:p.(Gly40AlafsTer7)	agrees	2	7	P/LP	False

Candidate	cDNA (MANE)	Predicted protein	Protein check	Papers	Identities	Authors' class	Only ClinVar-cited papers
G-lad1-078	NM_000211.5:c.106T>A	NP_000202.3:p.(Cys36Ser)	agrees	2	2	not stated	True
G-lad1-079	NM_000211.5:c.82_95del	NP_000202.3:p.(Phe28LeufsTer26)	agrees	1	1	not stated	False
G-lad1-080	NM_000211.5:c.77dup	NP_000202.3:p.(Lys27GlufsTer32)	agrees	1	1	not stated	True
G-lad1-081	NM_000211.5:c.66_67del	NP_000202.3:p.(Gln23GlyfsTer35)	agrees	1	2	not stated	True
G-lad1-082	NM_000211.5:c.49del	NP_000202.3:p.(Leu17SerfsTer33)	agrees	1	1	not stated	True
G-axd-001	NM_002055.5:c.*29C>T	NP_002046.1:p.(=)	no stated protein	1	1	not stated	False
G-axd-002	NM_002055.5:c.1292_1299delinsATC	NP_002046.1:p.(Val431AspfsTer14)	agrees	1	2	not stated	True
G-axd-003	NM_002055.5:c.1187C>T	NP_002046.1:p.(Thr396Ile)	agrees	2	4	P/LP	False
G-axd-004	NM_002055.5:c.1127G>C	NP_002046.1:p.(Arg376Pro)	agrees	1	1	P/LP	False
G-axd-005	NM_002055.5:c.1118A>G	NP_002046.1:p.(Glu373Gly)	agrees	1	1	P/LP	False
G-axd-006	NM_002055.5:c.1013T>C	NP_002046.1:p.(Leu338Pro)	agrees	1	1	P/LP	False
G-axd-007	NM_002055.5:c.778A>C	NP_002046.1:p.(Lys260Gln)	agrees	1	1	not stated	False
G-axd-008	NM_002055.5:c.740C>T	NP_002046.1:p.(Ser247Phe)	differs (stated pS247F, predicted S247F)	1	1	not stated	False
G-axd-009	NM_002055.5:c.619-1G>A	NP_002046.1:p.?	differs (stated E207del, predicted ?)	1	1	not stated	False
G-axd-010	NM_002055.5:c.616G>A	NP_002046.1:p.(Glu206Lys)	agrees	1	1	VUS	True
G-axd-011	NM_002055.5:c.363_371del	NP_002046.1:p.(Glu122_Arg124del)	agrees	1	1	VUS	False
G-axd-012	NM_002055.5:c.260T>A	NP_002046.1:p.(Val87Asp)	agrees	1	1	P/LP	False
G-axd-013	NM_002055.5:c.250A>T	NP_002046.1:p.(Ile84Phe)	agrees	1	1	P/LP	False
G-axd-014	NM_002055.5:c.219G>C	NP_002046.1:p.(Met73Ile)	agrees	2	1	not stated	False
G-axd-015	NM_002055.5:c.212C>A	NP_002046.1:p.(Ala71Glu)	no stated protein	1	1	VUS	False
G-axd-016	NM_002055.5:c.209G>C	NP_002046.1:p.(Arg70Pro)	agrees	1	1	P/LP	False
G-axd-017	NM_002055.5:c.54G>T	NP_002046.1:p.(Gly18=)	differs (stated p/G18V, predicted G18=)	1	1	P/LP	False
G-danon-001	NM_002294.3:c.1097_1098del	NP_002285.1:p.(Gln366ArgfsTer6)	no stated protein	1	1	not stated	True
G-danon-002	NM_002294.3:c.1093+2493G>C	NP_002285.1:p.?	differs (stated G384R, predicted ?)	1	4	not stated	False
G-danon-003	NM_002294.3:c.1079_1083del	NP_002285.1:p.(Gly360ValfsTer11)	agrees	1	2	P/LP	False
G-danon-004	NM_002294.3:c.1054del	NP_002285.1:p.(Val352PhefsTer6)	agrees	2	2	P/LP	False
G-danon-005	NM_002294.3:c.1039del	NP_002285.1:p.(Thr347ProfsTer4)	no stated protein	1	1	P/LP	False
G-danon-006	NM_002294.3:c.1009_1010del	NP_002285.1:p.(Val337PhefsTer12)	agrees	2	2	not stated	True
G-danon-007	NM_002294.3:c.966dup	NP_002285.1:p.(Ala323CysfsTer27)	agrees	1	1	not stated	False
G-danon-008	NM_002294.3:c.964del	NP_002285.1:p.(Asp322MetfsTer24)	agrees	1	2	not stated	False
G-danon-009	NM_002294.3:c.963G>A	NP_002285.1:p.(Trp321Ter)	no stated protein	2	2	not stated	False
G-danon-010	NM_002294.3:c.940del	NP_002285.1:p.(Ala314GlnfsTer32)	agrees	1	1	not stated	False
G-danon-011	NM_002294.3:c.865-2A>G	NP_002285.1:p.?	no stated protein	1	1	P/LP	False
G-danon-012	NM_002294.3:c.839del	NP_002285.1:p.(Lys280SerfsTer3)	no stated protein	1	1	not stated	True
G-danon-013	NM_002294.3:c.808dup	NP_002285.1:p.(Ala270GlyfsTer4)	agrees	1	1	not stated	False
G-danon-014	NM_002294.3:c.764_765insGA	NP_002285.1:p.(Asn255LysfsTer29)	no stated protein	1	1	P/LP	False
G-danon-015	NM_002294.3:c.750del	NP_002285.1:p.(Val251LeufsTer32)	agrees	1	1	P/LP	False

Candidate	cDNA (MANE)	Predicted protein	Protein check	Papers	Identities	Authors' class	Only ClinVar-cited papers
G-danon-016	NM_002294.3:c.749C>A	NP_002285.1:p.(Ser250Ter)	no stated protein	1	1	not stated	True
G-danon-017	NM_002294.3:c.741+2T>C	NP_002285.1:p.?	no stated protein	1	2	P/LP	False
G-danon-018	NM_002294.3:c.741+1G>T	NP_002285.1:p.?	no stated protein	1	3	P/LP	False
G-danon-019	NM_002294.3:c.733C>T	NP_002285.1:p.(Gln245Ter)	differs (stated G245*, predicted Q245*)	1	1	not stated	False
G-danon-020	NM_002294.3:c.645_651del	NP_002285.1:p.(Lys215AsnfsTer25)	agrees	1	1	P/LP	False
G-danon-021	NM_002294.3:c.573del	NP_002285.1:p.(Asp192ThrfsTer50)	agrees	2	1	not stated	True
G-danon-022	NM_002294.3:c.557-1G>C	NP_002285.1:p.?	no stated protein	1	1	not stated	False
G-danon-023	NM_002294.3:c.510_515delinsG	NP_002285.1:p.(Asp170GlufsTer15)	agrees	1	1	not stated	False
G-danon-024	NM_002294.3:c.470_471insTA	NP_002285.1:p.(Thr158LysfsTer26)	agrees	1	1	P/LP	False
G-danon-025	NM_002294.3:c.453del	NP_002285.1:p.(Phe151LeufsTer32)	agrees	3	3	P/LP	False
G-danon-026	NM_002294.3:c.427del	NP_002285.1:p.(Ile143SerfsTer5)	no stated protein	1	1	P/LP	False
G-danon-027	NM_002294.3:c.422T>A	NP_002285.1:p.(Leu141Ter)	no stated protein	1	1	not stated	True
G-danon-028	NM_002294.3:c.418del	NP_002285.1:p.(Leu140PhefsTer8)	agrees	3	3	P/LP	False
G-danon-029	NM_002294.3:c.406dup	NP_002285.1:p.(Thr136AsnfsTer3)	agrees	1	1	not stated	False
G-danon-030	NM_002294.3:c.388G>A	NP_002285.1:p.(Glu130Lys)	agrees	1	1	P/LP	False
G-danon-031	NM_002294.3:c.257_258del	NP_002285.1:p.(Pro86GlnfsTer26)	no stated protein	2	3	not stated	False
G-danon-032	NM_002294.3:c.103_104del	NP_002285.1:p.(Asp35PhefsTer20)	agrees	1	2	P/LP	False
G-danon-033	NM_002294.3:c.64+1G>A	NP_002285.1:p.?	no stated protein	1	2	not stated	False
G-danon-034	NM_002294.3:c.56T>C	NP_002285.1:p.(Leu19Pro)	no stated protein	1	1	not stated	True
G-danon-035	NM_002294.3:c.35C>A	NP_002285.1:p.(Ser12Ter)	agrees	2	2	P/LP	False

Reclassification candidates (53; exploratory). In ClinVar only as VUS or conflicting; 'strong' is a proposed rule.

Disease	cDNA (MANE)	Predicted protein	Papers	Identities	Authors' class	ClinVar class	ClinVar review status	Strong (proposed)
ha20	NM_001270508.2:c.65G>A	NP_001257437.1:p.(Arg22Gln)	1	1	VUS	VUS	criteria provided, single submitter	False
ha20	NM_001270508.2:c.386C>T	NP_001257437.1:p.(Thr129Met)	2	2	VUS	VUS	criteria provided, single submitter	True
ha20	NM_001270508.2:c.740C>T	NP_001257437.1:p.(Pro247Leu)	1	1	not stated	VUS	criteria provided, single submitter	False
ha20	NM_001270508.2:c.824T>C	NP_001257437.1:p.(Leu275Pro)	2	2	P/LP	VUS	criteria provided, single submitter	True
ha20	NM_001270508.2:c.838C>T	NP_001257437.1:p.(Arg280Trp)	1	1	not stated	conflicting	criteria provided, conflicting classifications	False
ha20	NM_001270508.2:c.929T>C	NP_001257437.1:p.(Ile310Thr)	2	3	VUS	VUS	criteria provided, multiple submitters, no conflicts	True
ha20	NM_001270508.2:c.1129G>A	NP_001257437.1:p.(Val377Met)	1	1	VUS	VUS	criteria provided, multiple submitters, no conflicts	False
ha20	NM_001270508.2:c.1504C>T	NP_001257437.1:p.(Arg502Trp)	2	1	not stated	VUS	criteria provided, single submitter	True
ha20	NM_001270508.2:c.1809del	NP_001257437.1:p.(Thr604ArgfsTer93)	3	3	not stated	VUS	criteria provided, single submitter	True

Disease	cDNA (MANE)	Predicted protein	Papers	Identities	Authors' class	ClinVar class	ClinVar review status	Strong (proposed)
ha20	NM_001270508.2:c.1906C>T	NP_001257437.1:p.(His636Tyr)	1	1	not stated	VUS	criteria provided, single submitter	False
ha20	NM_001270508.2:c.1939A>C	NP_001257437.1:p.(Thr647Pro)	5	9	mixed	conflicting	criteria provided, conflicting classifications	True
ha20	NM_001270508.2:c.2036T>C	NP_001257437.1:p.(Ile679Thr)	1	1	VUS	conflicting	criteria provided, conflicting classifications	False
ssadh	NM_001080.3:c.209T>C	NP_001071.1:p.(Leu70Pro)	1	3	VUS	VUS	criteria provided, single submitter	True
ssadh	NM_001080.3:c.269T>C	NP_001071.1:p.(Val90Ala)	1	1	VUS	VUS	criteria provided, single submitter	False
ssadh	NM_001080.3:c.434A>G	NP_001071.1:p.(Glu145Gly)	1	1	VUS	VUS	criteria provided, single submitter	False
ssadh	NM_001080.3:c.454G>C	NP_001071.1:p.(Glu152Gln)	1	2	VUS	VUS	criteria provided, single submitter	False
ssadh	NM_001080.3:c.479C>T	NP_001071.1:p.(Ser160Phe)	1	1	VUS	VUS	criteria provided, single submitter	False
ssadh	NM_001080.3:c.515G>A	NP_001071.1:p.(Arg172His)	3	3	mixed	conflicting	criteria provided, conflicting classifications	True
ssadh	NM_001080.3:c.517C>T	NP_001071.1:p.(Arg173Cys)	3	2	not stated	conflicting	criteria provided, conflicting classifications	True
ssadh	NM_001080.3:c.637C>T	NP_001071.1:p.(Arg213Trp)	1	1	P/LP	conflicting	criteria provided, conflicting classifications	False
ssadh	NM_001080.3:c.649G>A	NP_001071.1:p.(Ala217Thr)	1	1	not stated	conflicting	criteria provided, conflicting classifications	False
ssadh	NM_001080.3:c.923G>A	NP_001071.1:p.(Gly308Asp)	1	1	not stated	VUS	criteria provided, single submitter	False
ssadh	NM_001080.3:c.1015-3C>G	NP_001071.1:p.?	1	2	not stated	conflicting	criteria provided, conflicting classifications	False
ssadh	NM_001080.3:c.1480G>A	NP_001071.1:p.(Glu494Lys)	1	1	VUS	VUS	criteria provided, multiple submitters, no conflicts	False
ssadh	NM_001080.3:c.1501G>C	NP_001071.1:p.(Glu501Gln)	1	1	VUS	VUS	criteria provided, single submitter	False
ssadh	NM_001080.3:c.1529C>T	NP_001071.1:p.(Ser510Phe)	2	8	P/LP	conflicting	criteria provided, conflicting classifications	True
ssadh	NM_001080.3:c.1565A>G	NP_001071.1:p.(Asp522Gly)	1	1	P/LP	VUS	criteria provided, multiple submitters, no conflicts	False
tangier	NM_005502.4:c.5785G>T	NP_005493.2:p.(Val1929Phe)	1	1	not stated	VUS	criteria provided, single submitter	False
tangier	NM_005502.4:c.3626C>T	NP_005493.2:p.(Pro1209Leu)	1	1	VUS	VUS	criteria provided, single submitter	False
tangier	NM_005502.4:c.2879T>C	NP_005493.2:p.(Leu960Pro)	1	1	VUS	VUS	criteria provided, single submitter	False
tangier	NM_005502.4:c.1769G>T	NP_005493.2:p.(Trp590Leu)	1	1	P/LP	conflicting	criteria provided, conflicting classifications	False
tangier	NM_005502.4:c.1733C>G	NP_005493.2:p.(Pro578Arg)	1	1	VUS	VUS	criteria provided, multiple submitters, no conflicts	False
tangier	NM_005502.4:c.103A>G	NP_005493.2:p.(Ile35Val)	1	1	VUS	conflicting	criteria provided, conflicting classifications	False
lad1	NM_000211.5:c.1943C>T	NP_000202.3:p.(Pro648Leu)	1	1	VUS	VUS	criteria provided, multiple submitters, no conflicts	False
lad1	NM_000211.5:c.1358G>A	NP_000202.3:p.(Ser453Asn)	1	1	VUS	VUS	criteria provided, multiple submitters, no conflicts	False
lad1	NM_000211.5:c.758G>A	NP_000202.3:p.(Arg253His)	2	2	P/LP	VUS	criteria provided, single submitter	True
lad1	NM_000211.5:c.700G>A	NP_000202.3:p.(Glu234Lys)	1	2	P/LP	VUS	criteria provided, single submitter	False
lad1	NM_000211.5:c.400G>A	NP_000202.3:p.(Asp134Asn)	1	1	not stated	VUS	criteria provided, single submitter	False
lad1	NM_000211.5:c.392A>C	NP_000202.3:p.(Tyr131Ser)	1	1	not stated	VUS	criteria provided, single submitter	False
lad1	NM_000211.5:c.314T>C	NP_000202.3:p.(Leu105Pro)	3	5	P/LP	conflicting	criteria provided, conflicting classifications	True
axd	NM_002055.5:c.1171+473C>A	NP_002046.1:p.?	1	1	not stated	conflicting	criteria provided, conflicting classifications	False

Disease	cDNA (MANE)	Predicted protein	Papers	Identities	Authors' class	ClinVar class	ClinVar review status	Strong (proposed)
axd	NM_002055.5:c.1171+472G>A	NP_002046.1:p.?	2	5	P/LP	conflicting	criteria provided, conflicting classifications	True
axd	NM_002055.5:c.934G>T	NP_002046.1:p.(Glu312Ter)	1	1	not stated	VUS	criteria provided, single submitter	False
axd	NM_002055.5:c.667G>C	NP_002046.1:p.(Glu223Gln)	1	1	not stated	conflicting	criteria provided, conflicting classifications	False
axd	NM_002055.5:c.620A>T	NP_002046.1:p.(Glu207Val)	1	2	P/LP	conflicting	criteria provided, conflicting classifications	False
axd	NM_002055.5:c.584A>T	NP_002046.1:p.(Glu195Val)	1	1	VUS	VUS	criteria provided, single submitter	False
axd	NM_002055.5:c.197G>A	NP_002046.1:p.(Arg66Gln)	3	3	P/LP	conflicting	criteria provided, conflicting classifications	True
axd	NM_002055.5:c.173T>C	NP_002046.1:p.(Leu58Pro)	1	1	VUS	VUS	criteria provided, single submitter	False
danon	NM_002294.3:c.959A>G	NP_002285.1:p.(Tyr320Cys)	1	1	P/LP	conflicting	criteria provided, conflicting classifications	False
danon	NM_002294.3:c.928+3A>G	NP_002285.1:p.?	1	4	P/LP	conflicting	criteria provided, conflicting classifications	True
danon	NM_002294.3:c.815T>G	NP_002285.1:p.(Leu272Arg)	1	1	not stated	VUS	criteria provided, single submitter	False
danon	NM_002294.3:c.277G>A	NP_002285.1:p.(Gly93Arg)	1	1	not stated	conflicting	criteria provided, conflicting classifications	False
danon	NM_002294.3:c.52T>C	NP_002285.1:p.(Cys18Arg)	1	1	VUS	VUS	criteria provided, single submitter	False

Proposed same-person groups (unconfirmed; nothing merged).

Disease	Identities	n	Candidate
axd	AXD-ID0214 AXD-ID0430 AXD-ID0449 AXD-ID0450	4	G-axd-003
danon	DANON-ID0121 DANON-ID0320 DANON-ID0472	3	G-danon-028
danon	DANON-ID0137 DANON-ID0232	2	G-danon-031
danon	DANON-ID0196 DANON-ID0211	2	G-danon-009
danon	DANON-ID0252 DANON-ID0497	2	G-danon-035
danon	DANON-ID0409 DANON-ID0468	2	G-danon-025
ha20	HA20-ID0021 HA20-ID0136 HA20-ID0713 HA20-ID0714	4	G-ha20-016
ha20	HA20-ID0035 HA20-ID0036 HA20-ID0037 HA20-ID0317 HA20-ID0318 HA20-ID0319	6	G-ha20-002
ha20	HA20-ID0097 HA20-ID0100 HA20-ID0342 HA20-ID0618 HA20-ID0721	5	G-ha20-009
ha20	HA20-ID0099 HA20-ID0101	2	G-ha20-057
ha20	HA20-ID0201 HA20-ID0202 HA20-ID0205 HA20-ID0206 HA20-ID0630 HA20-ID0636 HA20-ID0637	7	G-ha20-043
ha20	HA20-ID0213 HA20-ID0734	2	G-ha20-046
ha20	HA20-ID0218 HA20-ID0584 HA20-ID0586 HA20-ID0709	4	G-ha20-001
ha20	HA20-ID0219 HA20-ID0730	2	G-ha20-055
ha20	HA20-ID0222 HA20-ID0764	2	G-ha20-034
ha20	HA20-ID0250 HA20-ID0752	2	G-ha20-015
ha20	HA20-ID0382 HA20-ID0433	2	G-ha20-021
ha20	HA20-ID0489 HA20-ID0594 HA20-ID0718	3	G-ha20-033
lad1	LAD1-ID0072 LAD1-ID0402	2	G-lad1-076

Supplementary Table S10. Memorization probe by disease and stratum

Model	Disease	Stratum	Papers	Specific / guess / none	Papers with a variant hit	Mean ROUGE-L F1	Longest run
Claude	AxD	after	1	0 / 1 / 0	0	0.194	4
Claude	AxD	older	17	0 / 12 / 5	0	0.203	4
Claude	AxD	recent	12	0 / 10 / 2	1	0.169	4
Claude	Danon	after	3	0 / 2 / 1	0	0.135	2
Claude	Danon	older	15	0 / 15 / 0	2	0.200	7
Claude	Danon	recent	12	0 / 10 / 2	1	0.192	4
Claude	HA20*	after	3	0 / 2 / 1	0	0.205	4

Model	Disease	Stratum	Papers	Specific / guess / none	Papers with a variant hit	Mean ROUGE-L F1	Longest run
Claude	HA20*	older	15	0 / 11 / 4	0	0.170	11
Claude	HA20*	recent	12	0 / 11 / 1	0	0.185	7
Claude	LAD-I	after	1	0 / 1 / 0	0	0.140	2
Claude	LAD-I	older	24	2 / 19 / 3	0	0.194	9
Claude	LAD-I	recent	5	0 / 5 / 0	0	0.258	6
Claude	SSADH	after	1	0 / 0 / 1	0	0.122	2
Claude	SSADH	older	25	2 / 17 / 6	0	0.185	5
Claude	SSADH	recent	4	0 / 2 / 2	0	0.192	4
Claude	Tangier	after	1	0 / 0 / 1	0	0.105	2
Claude	Tangier	older	23	0 / 21 / 2	0	0.189	5
Claude	Tangier	recent	6	0 / 6 / 0	0	0.192	5
DeepSeek	AxD	after	1	0 / 0 / 1	0	0.103	1
DeepSeek	AxD	older	17	0 / 0 / 17	0	0.156	4
DeepSeek	AxD	recent	12	0 / 0 / 12	0	0.150	4
DeepSeek	Danon	after	3	0 / 2 / 1	0	0.127	4
DeepSeek	Danon	older	15	0 / 15 / 0	2	0.145	3
DeepSeek	Danon	recent	12	0 / 10 / 2	1	0.146	3
DeepSeek	HA20*	after	3	0 / 0 / 3	0	0.152	3
DeepSeek	HA20*	older	15	0 / 0 / 15	0	0.154	7
DeepSeek	HA20*	recent	12	0 / 0 / 12	0	0.168	4
DeepSeek	LAD-I	after	1	0 / 0 / 1	0	0.170	2
DeepSeek	LAD-I	older	24	0 / 0 / 24	0	0.150	6
DeepSeek	LAD-I	recent	5	0 / 0 / 5	0	0.145	5
DeepSeek	SSADH	after	1	0 / 1 / 0	0	0.098	2
DeepSeek	SSADH	older	25	0 / 12 / 13	0	0.149	7
DeepSeek	SSADH	recent	4	0 / 1 / 3	0	0.173	6
DeepSeek	Tangier	after	1	0 / 0 / 1	0	0.157	3
DeepSeek	Tangier	older	23	0 / 0 / 23	0	0.137	4
DeepSeek	Tangier	recent	6	0 / 0 / 6	0	0.147	5

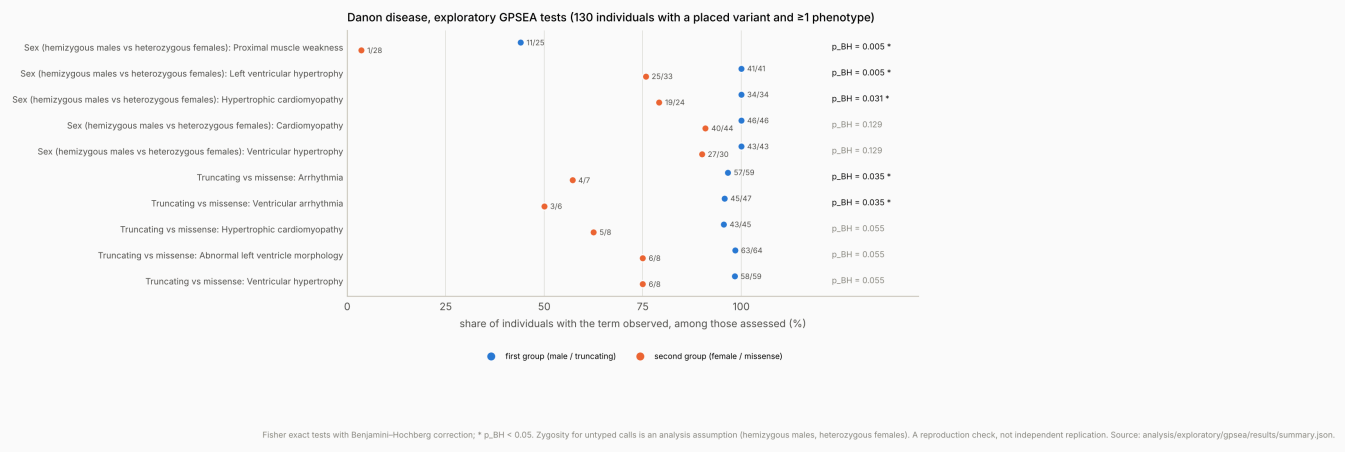
Supplementary Table S11. Open-weight run details

Disease	Papers	Harness version of final attempt	Papers with failed attempts	Retried calls	API calls	Prompt / completion tokens	Host-reported cost	Claims with an empty quote	...judged unsupported	...judged supported	Claims whose quote the span check did not find	Top unsupported fields
HA20*	85	v1 72; v2 10; v3 3	13	36	2792	348M / 15.8M	$10.25	1503	1185	236	48	`phenotype_checklist.status` 682; `individuals.is_proband` 477; `phenotypic_features.status` 79; `phenotypic_features.severity_or_detail` 69
SSADH	40	v2 39; v3 1	1	17	1419	221M / 8.7M	$5.54	591	438	144	8	`individuals.is_proband` 323; `phenotypic_features.status` 78; `phenotypic_features.severity_or_detail` 76; `variants.gene_symbol_as_stated` 42

Supplementary Table S12. GPSEA: the five lowest-p terms per test

Disease	Test	Term	Usable	Group shares	p	p_BH
HA20*	truncating vs missense	Fever	135	truncating 94/98; missense 33/37	0.214	1.000
HA20*	truncating vs missense	Recurrent fever	116	truncating 72/83; missense 26/33	0.393	1.000
HA20*	truncating vs missense	Oral ulcer	136	truncating 87/97; missense 34/39	0.763	1.000
HA20*	truncating vs missense	Arthritis	87	truncating 44/67; missense 14/20	0.793	1.000
HA20*	truncating vs missense	Erosion of oral mucosa	122	truncating 88/88; missense 34/34	1.000	1.000

Disease	Test	Term	Usable	Group shares	p	p_BH
AxD	hotspot vs other missense	Spasticity	34	hotspot 5/7; other missense 25/27	0.180	1.000
AxD	hotspot vs other missense	Upper motor neuron dysfunction	39	hotspot 7/7; other missense 31/32	1.000	1.000
AxD	hotspot vs other missense	Hypertonia	37	hotspot 6/6; other missense 30/31	1.000	1.000
AxD	hotspot vs other missense	Abnormal muscle tone	43	hotspot 8/8; other missense 34/35	1.000	1.000
AxD	hotspot vs other missense	Abnormal speech pattern	35	hotspot 5/5; other missense 29/30	1.000	1.000
Danon	truncating vs missense	Arrhythmia	66	truncating 57/59; missense 4/7	0.007	0.035
Danon	truncating vs missense	Ventricular arrhythmia	53	truncating 45/47; missense 3/6	0.008	0.035
Danon	truncating vs missense	Hypertrophic cardiomyopathy	53	truncating 43/45; missense 5/8	0.020	0.055
Danon	truncating vs missense	Abnormal left ventricle morphology	72	truncating 63/64; missense 6/8	0.031	0.055
Danon	truncating vs missense	Ventricular hypertrophy	67	truncating 58/59; missense 6/8	0.036	0.055
Danon	sex	Proximal muscle weakness	53	FEMALE 1/28; MALE 11/25	0.001	0.005
Danon	sex	Left ventricular hypertrophy	74	FEMALE 25/33; MALE 41/41	0.001	0.005
Danon	sex	Hypertrophic cardiomyopathy	58	FEMALE 19/24; MALE 34/34	0.009	0.031
Danon	sex	Cardiomyopathy	90	FEMALE 40/44; MALE 46/46	0.053	0.129
Danon	sex	Ventricular hypertrophy	73	FEMALE 27/30; MALE 43/43	0.065	0.129
SSADH	biallelic truncating	Seizure	41	truncating/truncating 4/7; truncating/other 7/10; other/other 17/24	0.808	1.000
SSADH	biallelic truncating	Neurodevelopmental abnormality	63	truncating/truncating 12/12; truncating/other 20/20; other/other 31/31	1.000	1.000
SSADH	biallelic truncating	Global developmental delay	48	truncating/truncating 12/12; truncating/other 15/15; other/other 21/21	1.000	1.000
SSADH	biallelic truncating	Neurodevelopmental delay	59	truncating/truncating 12/12; truncating/other 19/19; other/other 28/28	1.000	1.000
SSADH	biallelic truncating	Abnormal cerebral morphology	36	truncating/truncating 11/11; truncating/other 8/8; other/other 17/17	1.000	1.000
LAD-I	biallelic truncating	Increased inflammatory response	21	truncating/truncating 5/5; truncating/other 6/6; other/other 10/10	1.000	1.000
LAD-I	biallelic truncating	Increased total leukocyte count	29	truncating/truncating 6/6; truncating/other 11/11; other/other 12/12	1.000	1.000
LAD-I	biallelic truncating	Reduced cell surface marker level	17	truncating/truncating 5/5; truncating/other 3/3; other/other 9/9	1.000	1.000



Supplementary Figure S6. Danon disease GPSEA tests.

Supplementary Table S13. Spend segments (session cost readings)

Reading at	Work	Line (or split)	Amount
2026-10-18 20:28 UTC	Status review, DECISIONS/BUDGET, pre-registration draft, award details	unassigned	$19.63
2026-10-18 20:38 UTC	Pilot Stage 1 screening (838 abstracts)	1	$11.83
2026-10-18	Pilot Stage 2 smoke test (3 papers)	1	$12.56
2026-10-18	Between Stage 2 and Stage 3	unassigned	$1.91
2026-10-18	Pilot Stage 3: run-1, repeat, merge hand check	1	$194.43
2026-10-19	Pilot Stage 3b: preparation, run-2, repeat, merge hand check	1	$220.39

Reading at	Work	Line (or split)	Amount
2026-10-19	D-23/D-24 preparation (merge policy v4, input fixes, instructions v4)	4	$2.75
2026-10-19 03:02 UTC	v4 check (10 papers), Stage 4 HPO binding, Stage 5 gatekeeper (overlapping)	1:25,2:170	$116.81
2026-10-19	D-26 to D-28 work (repairs and scripts)	4	$12.71
2026-10-19	Gatekeeper v2 re-check on 50 audit claims	2	$17.17
2026-10-19	Between the re-check and D-29	unassigned	$4.78
2026-10-19	D-29 re-run (extraction v5, gatekeeper v3), Stage 6, re-audit (overlapping)	1:35.84,2:46.77	$58.92
2026-10-19 14:10 UTC	Items 1-5: corpus searches, ClinVar counts, coverage, drafts (before screening)	unassigned	$11.21
2026-10-20 02:48 UTC	Abstract screening, five diseases (reading froze; probably undercounted)	1	$4.64
2026-10-20 03:15 UTC	Full-text screening, five diseases	1	$27.39
2026-10-20 04:27 UTC	Between screening and the smoke run	unassigned	$7.09
2026-10-20 04:44 UTC	Pre-freeze smoke run (extraction v6 + gatekeeper v4)	1:70,2:30	$53.50
2026-10-20	Between the smoke run and the six-disease run	unassigned	$2.91
2026-10-20 22:25 UTC	Six-disease run: extraction, gatekeeper, repairs, 10% repeat, Stage 4 HPO binding	1:70,2:30	$832.70
2026-10-21 00:50 UTC	O1 sample, O2 concordance scripts, triage sample	unassigned	$14.70
2026-10-21 19:23 UTC	D-41: WS3 dossiers, WS5 probe (Claude), plus orchestration and D-10 harness work	3:64.10,5:5.34,unassigned:39.44	$108.88
2026-10-21 21:23 UTC	Before the open-weight gatekeeper	unassigned	$2.30
2026-10-24 15:55 UTC	Gatekeeper v4 on the open-weight runs (helper-token estimate $136.46, above this reading), probe orchestration, report, merge	2	$121.85

Supplementary Table S14. The development pilot's gates (HA20, from `reports/poc_report.md`)

Gate	Question	Threshold	Result
Q1	Open-access corpus	≥ 70% with full text	53.1% (85 of 160): not met
Q2	Verbatim provenance	every value has PMID + span	64,999 values, 100% verbatim: met
Q3	Valid phenopackets, normalized HGVS	100% valid; ≥ 95% cDNA normalized or flagged	749 of 749; 97.8%: met
Q4	Unsupported values	span match ≥ 98%; human ≤ 4% (n = 50)	100%; 6.0% and 6.3%: not met
Q5	Deduplication	0 false merges in the hand check	policy v3: 6 of 93: not met; v4: 0 of 50 (not independent)
Q6	Concordance harness	table with CIs vs dated snapshots	met (ClinVar, HPO)
Q7	Completeness	restricted recall ≥ 80%	9 of 9: met
Q8	Cost	six-disease projection ≤ $8,000	$1,676–4,189: met